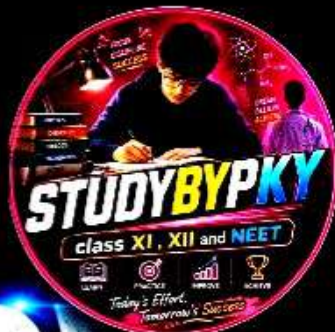
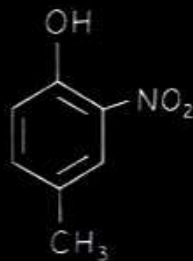


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CHAPTER 1

Classification of Elements and Periodicity in Properties

Modern Periodic Law, Nomenclature of Elements

- The IUPAC name of an element with atomic number 119 is. (2022)
 - ununcotium
 - ununennium
 - unnilennium
 - unununium
- Identify the incorrect match (2020)

	Name		IUPAC Official Name
(A)	Unnilunium	(i)	Mendelevium
(B)	Unniltrium	(ii)	Lawrencium
(C)	Unnilhexium	(iii)	Seaborgium
(D)	Unununium	(iv)	Darmstadtium

- (B), (ii)
- (C), (iii)
- (D), (iv)
- (A), (i)

Electronic Configurations and Types of Elements (s, p, d and f-Blocks)

- The element $Z = 114$ has been discovered recently. It will belong to which of the following family group and electronic configuration? (2017-Delhi)
 - Nitrogen family, $[\text{Rn}] 5f^{14}6d^{10}7s^27p^6$
 - Halogen family, $[\text{Rn}] 5f^{14}6d^{10}7s^27p^5$
 - Carbon family, $[\text{Rn}] 5f^{14}6d^{10}7s^27p^2$
 - Oxygen family, $[\text{Rn}] 5f^{14}6d^{10}7s^27p^4$
- The number of d-electrons in Fe^{2+} ($Z = 26$) is not equal to the number of electrons in which one of the following? (2015)
 - p-electrons in Cl ($Z = 17$)
 - d-electrons in Fe ($Z = 26$)
 - p-electrons in Ne ($Z = 10$)
 - s-electrons in Mg ($Z = 12$)

Periodic Trends in Properties of Elements

- From the following pairs of ions, which one is not an iso-electronic pair? (2021)
 - Na^+ , Mg^{2+}
 - Mn^{2+} , Fe^{3+}
 - Fe^{2+} , Mn^{2+}
 - O^{2-} , F^-
- For the second period elements, the correct increasing order of first ionisation enthalpy is: (2019)
 - $\text{Li} < \text{Be} < \text{B} < \text{C} < \text{N} < \text{O} < \text{F} < \text{Ne}$
 - $\text{Li} < \text{B} < \text{Be} < \text{C} < \text{O} < \text{N} < \text{F} < \text{Ne}$
 - $\text{Li} < \text{B} < \text{Be} < \text{C} < \text{N} < \text{O} < \text{F} < \text{Ne}$
 - $\text{Li} < \text{Be} < \text{B} < \text{C} < \text{O} < \text{N} < \text{F} < \text{Ne}$
- In which of the following options, the order of arrangement does not agree with the variation of property indicated against it? (2016 - I)
 - $\text{Li} < \text{Na} < \text{K} < \text{Rb}$ (increasing metallic radius)
 - $\text{Al}^{3+} < \text{Mg}^{2+} < \text{Na}^+ < \text{F}^-$ (increasing ionic size)
 - $\text{B} < \text{C} < \text{N} < \text{O}$ (increasing first ionization enthalpy)
 - $\text{I} < \text{Br} < \text{Cl} < \text{F}$ (increasing electron gain enthalpy)
- The formation of the oxide ion, $\text{O}^{2-}(\text{g})$ from oxygen atom requires first an exothermic step and then an endothermic step as shown below:

$$\text{O}(\text{g}) + \text{e}^- \rightarrow \text{O}^-(\text{g}); \Delta_f H^\circ = -141 \text{ kJ mol}^{-1}$$

$$\text{O}^-(\text{g}) + \text{e}^- \rightarrow \text{O}^{2-}(\text{g}); \Delta_f H^\circ = +780 \text{ kJ mol}^{-1}$$
 Thus, process of formation of O^{2-} in gas phase is unfavourable even though O^{2-} is isoelectronic with neon. It is due to the fact that, (2015 Re)
 - O^- ion has comparatively smaller size than oxygen atom
 - Oxygen is more electronegative
 - Addition of electron in oxygen results in larger size of the ion
 - Electron repulsion outweighs the stability gained by achieving noble gas configuration
- The species Ar , K^+ and Ca^{2+} contain the same number of electrons. In which order do their radii increase? (2015)
 - $\text{Ca}^{2+} < \text{Ar} < \text{K}^+$
 - $\text{Ca}^{2+} < \text{K}^+ < \text{Ar}$
 - $\text{K}^+ < \text{Ar} < \text{Ca}^{2+}$
 - $\text{Ar} < \text{K}^+ < \text{Ca}^{2+}$

10. Be^{2+} is isoelectronic with which of the following ions?
(2014)

- a. Li^+
c. Mg^{2+}
b. Na^+
d. H^+

11. Which of the following orders of ionic radii is correctly represented?
(2014)

- a. $\text{Na}^+ > \text{F}^- > \text{O}^{2-}$
c. $\text{Al}^{3+} > \text{Mg}^{2+} > \text{N}^{3-}$
b. $\text{O}^{2-} > \text{F}^- > \text{Na}^+$
d. $\text{H}^- > \text{H}^+ > \text{H}$

Answer Key

1	2	3	4	5	6	7	8	9	10	11
b	c	c	a	c	b	c,d	d	b	a	b

CHAPTER 2

The p-Block Elements

Group 13 Elements (Boron Family)

- The correct order of atomic radii in group 13 elements is (2018)
 - $B < Al < In < Ga < Tl$
 - $B < Al < Ga < In < Tl$
 - $B < Ga < Al < In < Tl$
 - $B < Ga < Al < Tl < In$
- Which one of the following elements is unable to form MF_3^- ion? (2018)
 - Ga
 - Al
 - In
 - B
- AlF_3 is soluble in HF only in presence of KF. It is due to the formation of: (2016-II)
 - $K_3[AlF_3H_3]$
 - $K_3[AlF_6]$
 - AlH_3
 - $K[AlF_3H_3]$
- The stability of +1 oxidation state among Al, Ga, In and Tl increases in the sequence: (2015 Re)
 - $In < Tl < Ga < Al$
 - $Ga < In < Al < Tl$
 - $Al < Ga < In < Tl$
 - $Tl < In < Ga < Al$

Compounds of Boron and Uses of B and Al and Their Compounds

- Which of the following statement is not correct about diborane? (2022)
 - Both the Boron atoms are sp^2 hybridised.
 - There are two 3-centre-2-electron bonds.
 - The four terminal B-H bonds are two centre two electron bonds
 - The four terminal Hydrogen atoms and the two Boron atoms lie in one plane.
- Boric acid is an acid because its molecule: (2016 - II)
 - Accepts OH^- from water releasing proton
 - Combines with proton from water molecule
 - Contains replaceable H^+ ion
 - Gives up a proton

Group 14 Elements (Carbon Family)

- Match the following : (2020)

Oxide	Nature
A. CO	(i) Basic
B. BaO	(ii) Neutral
C. Al_2O_3	(iii) Acidic
D. Cl_2O_7	(iv) Amphoteric

 Which of the following is correct option?
 (A) (B) (C) (D)
 a. (ii) (i) (iv) (iii)
 b. (iii) (iv) (i) (ii)
 c. (iv) (iii) (ii) (i)
 d. (i) (ii) (iii) (iv)
- Which of the following oxide is amphoteric in nature? (2020-Covid)
 - SiO_2
 - GeO_2
 - CO_2
 - SnO_2
- Which of the following species is not stable? (2019)
 - $[SiF_6]^{2-}$
 - $[GeCl_6]^{2-}$
 - $[Sn(OH)_6]^{2-}$
 - $[SiCl_6]^{2-}$
- Which of the following is incorrect statement? (2019)
 - PbF_4 is covalent in nature
 - $SiCl_4$ is easily hydrolysed
 - GeX_4 ($X = F, Cl, Br, I$) is more stable than GeX_2
 - SnF_4 is ionic in nature
- It is because of inability of ns^2 electrons of the valence shell to participate in bonding that: (2017-Delhi)
 - Sn^{4+} is reducing while Pb^{4+} is oxidising
 - Sn^{2+} is reducing while Pb^{4+} is oxidising
 - Sn^{2+} is oxidising while Pb^{4+} is reducing
 - Sn^{2+} and Pb^{2+} are both oxidising and reducing

Allotropes of Carbon

- Choose the correct statement: (2022)
 - Both diamond and graphite are used as dry lubricants.
 - Diamond and graphite have two dimensional network
 - Diamond is covalent and graphite is ionic
 - Diamond is sp^3 hybridised and graphite is sp^2 hybridized.

13. Which of the following structure is similar to graphite?
(2013)

- a. BN b. B
c. B₄C d. B₂H₆

Some Important Compounds of C & Si

14. Identify the correct statements from the following : (2020)

- (1) CO₂(g) is used as refrigerant for ice-cream and frozen food.
(2) The structure of C₆₀ contains twelve six carbon rings and twenty five carbon rings.
(3) ZSM-5, a type of zeolite, is used to convert alcohols into gasoline.
(4) CO is colorless and odourless gas.

- a. (1) and (3) only b. (2) and (3) only
c. (3) and (4) only d. (1), (2) and (3) only

15. Which of the following is not correct about carbon monoxide?
(2020)

- a. It reduces oxygen carrying ability of blood.
b. The carboxyhaemoglobin (haemoglobin bound to CO) is less stable than oxyhaemoglobin.
c. It is produced due to incomplete combustion.
d. It forms carboxyhaemoglobin.

16. Which of these is not a monomer for a high molecular mass silicone polymer?
(2013)

- a. MeSiCl₃ b. Me₂SiCl₂
c. Me₃SiCl d. PhSiCl₃

17. The basic structural unit of silicates is: (2013)

- a. SiO⁻ b. SiO₄⁴⁻
c. SiO₃²⁻ d. SiO₄²⁻

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
c	d	b	c	a	a	a	d	d	a	b	d	a	c	b	c	b

CHAPTER 3

The p-Block Elements (Group 15 to 18)

Group 15 Elements (Nitrogen Family)

1. Urea reacts with water to form A which will decompose to form B. B when passed through Cu^{2+} (aq), deep blue colour solution C is formed. What is the formula of C from the following? (2020)

a. $[\text{Cu}(\text{NH}_3)_4]^{2+}$ b. $\text{Cu}(\text{OH})_2$
c. $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$ d. CuSO_4

2. Among the compounds shown below which one revealed a linear structure? (2020-Covid)

a. HOCl b. O_3
c. N_2O d. NO_2

3. Match the following: (2019)

Column-I	Column-II
A. Pure nitrogen	i. Chlorine
B. Haber process	ii. Sulphuric acid
C. Contact process	iii. Ammonia
D. Deacon's process	iv. Sodium azide or Barium azide

Which of the following is the correct option?

A	B	C	D
a. i	ii	iii	iv
b. ii	iv	i	iii
c. iii	iv	ii	i
d. iv	iii	ii	i

4. The correct order of N-compounds in its decreasing order of oxidation states is (2018)

a. HNO_3 , NO , N_2 , NH_4Cl b. HNO_3 , NO , NH_4Cl , N_2
c. NH_4Cl , N_2 , NO , HNO_3 d. HNO_3 , NH_4Cl , NO , N_2

5. Which is the correct statement for the given acids? (2016 - I)

a. Phosphinic acid is a diprotic acid while phosphonic acid is a monoprotic acid.
b. Phosphinic acid is a monoprotic acid while phosphonic acid is a diprotic acid
c. Both are diprotic acids
d. Both are triprotic acids

6. When copper is heated with conc. HNO_3 it produces: (2016 - I)

a. $\text{Cu}(\text{NO}_3)_2$ and N_2O b. $\text{Cu}(\text{NO}_3)_2$ and NO_2
c. $\text{Cu}(\text{NO}_3)_2$ and NO d. $\text{Cu}(\text{NO}_3)_2$, NO and NO_2

7. Strong reducing behaviour of H_3PO_2 is due to: (2015 Re)

a. Presence of two $-\text{OH}$ groups and one $\text{P}-\text{H}$ bond
b. Presence of one $-\text{OH}$ group and two $\text{P}-\text{H}$ bonds
c. High electron gain enthalpy of phosphorus
d. High oxidation state of phosphorus

8. Nitrogen dioxide and sulphur dioxide have some properties in common. Which property is shown by one of these compounds, but not by the other? (2015)

a. Is soluble in water b. Is used as a food preservative
c. Forms 'acid-rain' d. Is a reducing agent

9. Which of the following does not give oxygen on heating? (2013)

a. KClO_3 b. $\text{Zn}(\text{ClO}_3)_2$
c. $\text{K}_2\text{Cr}_2\text{O}_7$ d. $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$

Group 16 Elements (Oxygen Family)

10. Given below are two statements: (2022)

Statement I : The boiling points of the following hydrides of group 16 elements increases in the order-



Statement II : The boiling points of these hydrides increase with increase in molar mass.

In the light of the above statements, choose the most appropriate answer from the options given below:

a. Statements I is incorrect but Statements II is correct.
b. Both Statements I and Statements II are correct
c. Both Statements I and Statements II are incorrect
d. Statements I is correct but Statements II is incorrect

11. In which one of the following arrangements the given sequence is not strictly according to the properties indicated against it? (2021)

a. $\text{H}_2\text{O} < \text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te}$: Increasing pK_a values
b. $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3 < \text{SbH}_3$: Increasing acidic character
c. $\text{CO}_2 < \text{SiO}_2 < \text{SnO}_2 < \text{PbO}_2$: Increasing oxidizing power
d. $\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$: Increasing acidic strength

12. Which of the following oxoacid of sulphur has – O – O – linkage? (2020)

- H_2SO_4 , sulphuric acid
- $\text{H}_2\text{S}_2\text{O}_8$, peroxodisulphuric acid
- $\text{H}_2\text{S}_2\text{O}_7$, pyrosulphuric acid
- H_2SO_3 , sulphurous acid

13. The reaction of concentrated sulphuric acid with carbohydrates ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) is an example of (2020-Covid)

- Oxidation
- Reduction
- Sulphonation
- Dehydration

14. Which is the correct thermal stability order for H_2E (E = O, S, Se, Te and Po)? (2019)

- $\text{H}_2\text{S} < \text{H}_2\text{O} < \text{H}_2\text{Se} < \text{H}_2\text{Te} < \text{H}_2\text{Po}$
- $\text{H}_2\text{O} < \text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te} < \text{H}_2\text{Po}$
- $\text{H}_2\text{Po} < \text{H}_2\text{Te} < \text{H}_2\text{Se} < \text{H}_2\text{S} < \text{H}_2\text{O}$
- $\text{H}_2\text{Se} < \text{H}_2\text{Te} < \text{H}_2\text{Po} < \text{H}_2\text{O} < \text{H}_2\text{S}$

15. In which pair of ions both the species contain S – S bond? (2017-Delhi)

- $\text{S}_4\text{O}_6^{2-}, \text{S}_2\text{O}_7^{2-}$
- $\text{S}_2\text{O}_7^{2-}, \text{S}_2\text{O}_3^{2-}$
- $\text{S}_4\text{O}_6^{2-}, \text{S}_2\text{O}_3^{2-}$
- $\text{S}_2\text{O}_7^{2-}, \text{S}_2\text{O}_8^{2-}$

16. Which of the statements given below is incorrect? (2015 Re)

- O_3 molecule is bent
- ONF is isoelectronic with O_2N^-
- OF_2 is an oxide of fluorine
- Cl_2O_7 is an anhydride of perchloric acid

17. Acidity of diprotic acids in aqueous solutions increases in the order: (2014)

- $\text{H}_2\text{Se} < \text{H}_2\text{S} < \text{H}_2\text{Te}$
- $\text{H}_2\text{Te} < \text{H}_2\text{S} < \text{H}_2\text{Se}$
- $\text{H}_2\text{Se} < \text{H}_2\text{Te} < \text{H}_2\text{S}$
- $\text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te}$

Group 17 Elements (Halogens)

18. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R) (2022)

Assertion (A): ICl is more reactive than I_2 .

Reason (R): I-Cl bond is weaker than I-I bond

In the light of the above statements, choose the most appropriate answer from the options given below:

- (A) is not correct but (R) is correct.
- Both (A) and (R) are correct and (R) is the correct explanation of (A).
- Both (A) and (R) are correct but (R) is not the correct explanation of (A).
- (A) is correct but (R) is not correct.

19. Statement-I: Acid strength increases in the order given as $\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$. (2021)

Statement II: As the size of the elements F, Cl, Br, I increases down the group, the bond strength of HF, HCl, HBr and HI decreases and so the acid strength increases.

In the light of the above statements, choose the correct answer from the options given below.

- Both statement I and statement II are false
- Statement I is correct but statement II is false
- Statement I is incorrect but Statement II is true
- Both statement I and statement II are true

20. Which one of the following reactions does not come under hydrolysis type reaction? (2020-Covid)

- $\text{Li}_3\text{N}(\text{s}) + 3\text{H}_2\text{O}(\text{l}) \rightarrow \text{NH}_3(\text{g}) + 3\text{LiOH}(\text{aq})$
- $2\text{F}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow 4\text{HF}(\text{aq}) + \text{O}_2(\text{g})$
- $\text{P}_4\text{O}_{10}(\text{s}) + 6\text{H}_2\text{O}(\text{l}) \rightarrow 4\text{H}_3\text{PO}_4(\text{aq})$
- $\text{SiCl}_4(\text{l}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow \text{SiO}_2(\text{s}) + 4\text{HCl}(\text{aq})$

21. Which of the following statements is not true for halogens? (2018)

- All form monobasic oxyacids
- All are oxidizing agents
- Chlorine has the highest electron-gain enthalpy
- All but fluorine show positive oxidation states

22. In the structure of ClF_3 , the number of lone pair of electrons on central atom 'Cl' is (2018)

- One
- Two
- Three
- Four

23. Match the interhalogen compounds of Column-I with the geometry in Column-II and assign the correct code: (2017-Delhi)

Column-I		Column-II	
(A)	XX'	(i)	T-shape
(B)	XX'_3	(ii)	Pentagonal bipyramidal
(C)	XX'_5	(iii)	Linear
(D)	XX'_7	(iv)	Square-pyramidal
		(v)	Tetrahedral

Code:

- | | | | |
|----------|-------|-------|------|
| (A) | (B) | (C) | (D) |
| a. (iv) | (iii) | (ii) | (i) |
| b. (iii) | (iv) | (i) | (ii) |
| c. (iii) | (i) | (iv) | (ii) |
| d. (v) | (iv) | (iii) | (ii) |

24. Which one of the following orders is correct for the bond dissociation enthalpy of halogen molecules? (2016 - I)

- $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$
- $\text{I}_2 > \text{Br}_2 > \text{Cl}_2 > \text{F}_2$
- $\text{Cl}_2 > \text{Br}_2 > \text{F}_2 > \text{I}_2$
- $\text{Br}_2 > \text{I}_2 > \text{F}_2 > \text{Cl}_2$

25. Among the following, the correct order of acidity is:

(2016 - I)

- $\text{HClO}_4 < \text{HClO}_2 < \text{HClO} < \text{HClO}_3$
- $\text{HClO}_3 < \text{HClO}_4 < \text{HClO}_2 < \text{HClO}$
- $\text{HClO} < \text{HClO}_2 < \text{HClO}_3 < \text{HClO}_4$
- $\text{HClO}_2 < \text{HClO} < \text{HClO}_3 < \text{HClO}_4$

26. The variation of the boiling points of the hydrogen halides is in the order $\text{HF} > \text{HI} > \text{HBr} > \text{HCl}$. What explains the higher boiling point of hydrogen fluoride? (2015 Re)

- The effect of nuclear shielding is much reduced in fluorine which polarizes the HF molecule
- The electronegativity of fluorine is much higher than for other elements in the group
- There is strong hydrogen bonding between HF molecules
- The bond energy of HF molecules is greater than in other hydrogen halides

27. Which is the strongest acid in the following? (2013)

- H_2SO_4
- HClO_3
- HClO_4
- H_2SO_3

Group 18 Elements (Noble Gases)

28. Noble gases are named because of their inertness towards reactivity. Identify an incorrect statement about them. (2021)

- Noble gases have very high melting and boiling points
- Noble gases have weak dispersion forces
- Noble gases have large positive values of electron gain enthalpy
- Noble gases are sparingly soluble in water

29. Match the compounds of Xe in Column-I with the molecular structure in Column-II. (2020-Covid)

Column-I		Column-II	
(A)	XeF_2	(i)	Square planar
(B)	XeF_4	(ii)	Linear
(C)	XeO_3	(iii)	Square pyramidal
(D)	XeOF_4	(iv)	Pyramidal

- A-ii B-i C-iv D-iii
- A-ii B-i C-iii D-iv
- A-ii B-iv C-iii D-i
- A-ii B-iii C-i D-iv

30. Match the Xenon compounds in Column-I with its structure in Column-II and assign the correct code: (2019)

Column-I		Column-II	
(A)	XeF_4	(i)	Pyramidal
(B)	XeF_6	(ii)	Square planar
(C)	XeOF_4	(iii)	Distorted octahedral
(D)	XeO_3	(iv)	Square pyramidal

Code:

- | | | | |
|--------|-----|-----|----|
| A | B | C | D |
| a. i | ii | iii | iv |
| b. ii | iii | iv | i |
| c. ii | iii | i | iv |
| d. iii | iv | i | ii |

31. Which of the following pairs of compounds is isoelectronic and isostructural? (2017-Delhi)

- | | |
|------------------------------------|--------------------------------------|
| a. IF_3 , XeF_2 | b. BeCl_2 , XeF_2 |
| c. TeI_2 , XeF_2 | d. IBr_2^- , XeF_2 |

32. The correct geometry and hybridization for XeF_4 are: (2016 - II)

- | | |
|---|--|
| a. Planar triangle, sp^3d^3 | b. Square planar, sp^3d^2 |
| c. Octahedral, sp^3d^2 | d. Trigonal bipyramidal, sp^3d |

33. Match the compound given in Column-I with the hybridization and shape given in Column-II and mark the correct option. (2016 - I)

Column-I		Column-II	
(A)	XeF_6	(i)	Distorted octahedral
(B)	XeO_3	(ii)	Square planar
(C)	XeOF_4	(iii)	Pyramidal
(D)	XeF_4	(iv)	Square pyramidal

Code:

- | | | | |
|---------|-------|------|-------|
| (A) | (B) | (C) | (D) |
| a. (iv) | (i) | (ii) | (iii) |
| b. (i) | (iii) | (iv) | (ii) |
| c. (i) | (ii) | (iv) | (iii) |
| d. (iv) | (iii) | (i) | (ii) |

34. In which of the following pairs, both the species are not isostructural? (2015 Re)

- | | |
|------------------------------------|---------------------------------------|
| a. XeF_4 , XeO_4 | b. SiCl_4 , PCl_4^+ |
| c. Diamond, silicon carbide | d. NH_3 , PH_3 |

35. XeF_2 is isostructural with: (2013)

- | | |
|--------------------|---------------------|
| a. TeF_2 | b. ICl_2^- |
| c. SbCl_3 | d. BaCl_2 |

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
a	c	d	a	b	b	b	b	d	c	a	b	d	c	c	c	d
18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
b	d	b	d	b	c	c	c	c	c	a	a	b	d	b	b	a
35																
b																

CHAPTER 4

The d- and f-Block Elements

Transition Elements (d-Block)

- Zr ($Z = 40$) and Hf ($Z = 72$) have similar atomic and ionic radii because of: (2021)
 - Diagonal relationship
 - Lanthanoid contraction
 - Having similar chemical properties
 - Belonging to same group
- The calculated spin only magnetic moment of Cr^{2+} ion is (2020)
 - 4.90 BM
 - 5.92 BM
 - 2.84 BM
 - 3.87 BM
- Match the following aspects with the respective metal. (2020-Covid)

Aspects		Metal	
A.	The metal which reveals a maximum number of oxidation states	i.	Scandium
B.	The metal although placed in 3d block is considered not as a transition element	ii.	Copper
C.	The metal which does not exhibit variable oxidation states	iii.	Manganese
D.	The metal which in +1 oxidation state in aqueous solution undergoes disproportionation	iv.	Zinc

Select the correct option:

- A-iii B-iv C-i D-ii
- A-iii B-i C-iv D-ii
- A-ii B-iv C-i D-iii
- A-i B-iv C-ii D-iii

- Match the metal ions given in Column I with the spin magnetic moments of the ions given in Column II and assign the correct code: (2018)

Column-I		Column-II	
A.	Co^{3+}	i.	$\sqrt{8}$ B.M
B.	Cr^{3+}	ii.	$\sqrt{35}$ B.M
C.	Fe^{3+}	iii.	$\sqrt{3}$ B.M
D.	Ni^{2+}	iv.	$\sqrt{24}$ B.M
		v.	$\sqrt{15}$ B.M

- A-iv B-v C-ii D-i
 - A-i B-ii C-iii D-iv
 - A-iii B-v C-i D-ii
 - A-iv B-i C-ii D-iii
- HgCl_2 and I_2 both when dissolved in water containing I^- ions the pair of species formed is: (2017-Delhi)
 - $\text{Hg}_2\text{I}_2, \text{I}^-$
 - $\text{HgI}_2, \text{I}_3^-$
 - HgI_2, I^-
 - $\text{HgI}_4^{2-}, \text{I}_3^-$
 - Which of the following processes does not involve oxidation of iron? (2015)
 - Decolourisation of blue CuSO_4 solution by iron
 - Formation of $\text{Fe}(\text{CO})_5$ from Fe
 - Liberation of H_2 from steam by iron at high temperature
 - Rusting of iron sheets
 - Magnetic moment 2.84 B.M. is given by: (Atomic numbers, Ni = 28, Ti = 22, Cr = 24, Co = 27) (2015)
 - Ti^{3+}
 - Cr^{2+}
 - Co^{2+}
 - Ni^{2+}
 - Because of lanthanoid contraction, which of the following pairs of elements have nearly same atomic radii? (Numbers in the parenthesis are atomic numbers) (2015)
 - Zr (40) and Nb (41)
 - Zr (40) and Hf (72)
 - Zr (40) and Ta (73)
 - Ti (22) and Zr (40)
 - Magnetic moment 2.83 BM is given by which of the following ions? (2014)

(Atomic Number Ti = 22, Cr = 24, Mn = 25, Ni = 28)

 - Ni^{2+}
 - Cr^{3+}
 - Mn^{2+}
 - Ti^{3+}

10. Which of the following statements about the interstitial compounds is incorrect? (2013)
- They retain metallic conductivity
 - They are chemically reactive
 - They are much harder than the pure metal
 - They have higher melting points than the pure metal

Compounds of Transition Elements

11. In the neutral or faintly alkaline medium, KMnO_4 oxidises iodide into iodate. The change in oxidation state of manganese in this reaction is from (2022)
- + 6 to + 5
 - + 7 to + 4
 - + 6 to + 4
 - + 7 to + 3
12. Identify the incorrect statement. (2020)
- The transition metals and their compounds are known for their catalytic activity due to their ability to adopt multiple oxidation states and to form complexes.
 - Interstitial compounds are those that are formed when small atoms like H, C or N are trapped inside the crystal lattices of metals.
 - The oxidation states of chromium in CrO_4^{2-} and $\text{Cr}_2\text{O}_7^{2-}$ are not the same.
 - $\text{Cr}^{2+}(\text{d}^4)$ is a stronger reducing agent than $\text{Fe}^{2+}(\text{d}^6)$ in water.
13. The manganate and permanganate ions are tetrahedral, due to: (2019)
- The p-bonding involves overlap of p-orbitals of oxygen with d-orbitals of manganese
 - There is no p-bonding
 - The p-bonding involves overlap of p-orbitals of oxygen with p-orbitals of manganese
 - The p-bonding involves overlap of d-orbitals of oxygen with d-orbitals of manganese
14. Which one of the following ions exhibits d-d transition and paramagnetism as well? (2018)
- CrO_4^{2-}
 - $\text{Cr}_2\text{O}_7^{2-}$
 - MnO_4^{2-}
 - MnO_4^-
15. Name the gas that can readily decolourise acidified KMnO_4 solution: (2017-Delhi)
- P_2O_5
 - CO_2
 - SO_2
 - NO_2
16. Which one of the following statements is correct when SO_2 is passed through acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution? (2016-I)
- SO_2 is reduced
 - Green $\text{Cr}_2(\text{SO}_4)_3$ is formed
 - The solution turns blue
 - The solution is decolourised
17. Assuming complete ionisation, same moles of which of the following compounds will require the least amount of acidified KMnO_4 for complete oxidation? (2015 Re)
- $\text{Fe}(\text{NO}_2)_2$
 - FeSO_4
 - FeSO_3
 - FeC_2O_4

Inner Transition Elements (Lanthanoids and Actinoids)

18. Gadolinium has a **low** value of third ionisation enthalpy because of (2022)
- High basic character
 - Small size
 - High exchange enthalpy
 - High electronegativity
19. The incorrect statement among the following is: (2021)
- Most of the trivalent Lanthanoid ions are colorless in the solid state
 - Lanthanoids are good conductors of heat and electricity
 - Actinoids are highly reactive metals, especially when finely divided
 - Actinoid contraction is greater for element to element than Lanthanoid contraction
20. Identify the incorrect statement from the following: (2020-Covid)
- Lanthanoids reveal only +3 oxidation state.
 - The lanthanoid ions other than the f^0 type and the f^{14} type are all paramagnetic.
 - The overall decreases in atomic and ionic radii from lanthanum to lutetium is called lanthanoid contraction.
 - Zirconium and Hafnium have identical radii of 160 pm and 159 pm, respectively as a consequence of lanthanoid contraction.
21. Match the element in Column-I with that in Column-II. (2020-Covid)

Column-I		Column-II	
A.	Copper	i.	Non-metal
B.	Fluorine	ii.	Transition metal
C.	Silicon	iii.	Lanthanoid
D.	Cerium	iv.	Metalloid

Select the correct option:

- A-i B-ii C-iii D-iv
 - A-ii B-iv C-i D-iii
 - A-ii B-i C-iv D-iii
 - A-iv B-iii C-i D-ii
22. The reason for greater range of oxidation states in actinoids is attributed to: (2017-Delhi)
- 4f and 5d levels being close in energies
 - The radioactive nature of actinoids
 - Actinoid contraction
 - 5f, 6d and 7s levels having comparable energies
23. Which one of the following statements related to lanthanons is incorrect? (2016-II)
- All the lanthanons are much more reactive than aluminium
 - $\text{Ce}(+4)$ solutions are widely used as oxidizing agent in volumetric analysis
 - Europium shows +2 oxidation state.
 - The basicity decreases as the ionic radius decreases from Pr to Lu.

24. The electronic configurations of Eu (Atomic Number 63) Gd (Atomic Number 64) and Tb (Atomic Number 65) are: (2016-I)
- $[\text{Xe}]4f^7 6s^2$, $[\text{Xe}]4f^7 5d^1 6s^2$ and $[\text{Xe}]4f^9 6s^2$
 - $[\text{Xe}]4f^7 6s^2$, $[\text{Xe}]4f^8 6s^2$ and $[\text{Xe}]4f^8 5d^1 6s^2$
 - $[\text{Xe}]4f^6 5d^1 6s^2$, $[\text{Xe}]4f^7 5d^1 6s^2$ and $[\text{Xe}]4f^9 5d^1 6s^2$
 - $[\text{Xe}]4f^6 5d^1 6s^2$, $[\text{Xe}]4f^7 5d^1 6s^2$ and $[\text{Xe}]4f^8 5d^1 6s^2$
25. Gadolinium belongs to 4f series. Its atomic number is 64. Which of the following is the correct electronic configuration of gadolinium? (2015 Re)
- $[\text{Xe}]4f^6 5d^2 6s^2$
 - $[\text{Xe}]4f^8 6d^2$
 - $[\text{Xe}]4f^9 5s^1$
 - $[\text{Xe}]4f^7 5d^1 6s^2$
26. Reason of lanthanoid contraction is: (2014)
- Increasing nuclear charge
 - Decreasing nuclear charge
 - Decreasing screening effect
 - Negligible screening effect of 'f' orbitals
27. Which of the following lanthanoid ions is diamagnetic? (Atomic Number Ce = 58, Sm = 62, Eu = 63, Yb = 70) (2013)
- Ce^{2+}
 - Sm^{2+}
 - Eu^{2+}
 - Yb^{2+}

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
b	a	a	a	d	b	d	b	a	b	b	c	a	c	c	b	b
18	19	20	21	22	23	24	25	26	27							
c	a	a	c	d	a	a	d	d	d							

CHAPTER 5

Chemical Bonding and Molecular Structure

Covalent Bond, Electrovalent Bond, Lattice Enthalpy and Octet Rule

- Which of the following species contains equal number of σ and π -bonds? (2015 Re)
 - (CN)₂
 - (CH)₂(CN)₂
 - HCO₃⁻
 - XeO₄
- Which one of the following molecules contains no π bond? (2013)
 - SO₂
 - NO₂
 - CO₂
 - H₂O
- Which of the following is electron-deficient? (2013)
 - (CH₃)₂
 - (SiH₃)₂
 - (BH₃)₂
 - PH₃

Bond Parameters and Dipole Moment

- Which of the following molecules is non-polar in nature? (2021)
 - CH₂O
 - SbCl₅
 - NO₂
 - POCl₃
- Which of the following set of molecules will have zero dipole moment? (2020)
 - Boron trifluoride, hydrogen fluoride, carbon dioxide, 1,3-dichlorobenzene
 - Nitrogen trifluoride, beryllium difluoride, water, 1,3-dichlorobenzene
 - Boron trifluoride, beryllium difluoride, carbon dioxide, 1,4-dichlorobenzene
 - Ammonia, beryllium difluoride, water, 1,4-dichlorobenzene
- Which one of the following pair of species have the same bond order? (2017-Delhi)
 - N₂, O₂⁻
 - CO, NO
 - O₂, NO⁺
 - CN⁻, CO
- Consider the molecules CH₄, NH₃ and H₂O. Which of the given statement is false? (2016 - I)
 - The H—C—H bond angle in CH₄ is larger than the H—N—H bond angle in NH₃
 - The H—C—H bond angle in CH₄, the H—N—H bond angle in NH₃, and the H—O—H bond angle in H₂O are all greater than 90°.
 - Then H—O—H bond angle in H₂O is larger than the H—C—H bond angle in CH₄
 - The H—O—H bond angle in H₂O is smaller than the H—N—H bond angle in NH₃

- Which of the following molecules has the maximum dipole moment? (2014)
 - CH₄
 - NH₃
 - NF₃
 - CO₂

VSEPR Theory

- Amongst the following, which one will have maximum 'lone pair-lone pair' electron repulsions? (2022)
 - XeF₂
 - ClF₃
 - IF₅
 - SF₄
- Match List-I with List-II.

List-I		List-II	
(A)	PCl ₅	(i)	Square pyramidal
(B)	SF ₆	(ii)	Trigonal planar
(C)	BrF ₅	(iii)	Octahedral
(D)	BF ₃	(iv)	Trigonal bipyramidal

Choose the correct answer from the options given below. (2021)

- A-ii B-iii C-iv D-i
 - A-iii B-i C-iv D-ii
 - A-iv B-iii C-ii D-i
 - A-iv B-iii C-i D-ii
- Identify the wrongly matched pair. (2020-Covid)

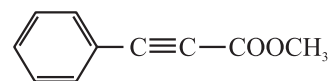
Molecule	Shape or geometry of molecule
a. SF ₆	Octahedral
b. BeCl ₂	Linear
c. NH ₃	Trigonal pyramidal
d. PCl ₅	Trigonal planar

12. Identify the incorrect statement related to PCl_5 from the following: (2019)
- Three equatorial P-Cl bonds make an angle of 120° with each other
 - Two axial P-Cl bonds make an angle of 180° with each other
 - Axial P-Cl bonds are longer than equatorial P-Cl bonds
 - PCl_5 molecule is non-reactive
13. The species, having bond angles of 120° is (2017-Delhi)
- BCl_3
 - PH_3
 - ClF_3
 - NCl_3
14. Among the following which one is a wrong statement? (2016 - II)
- SeF_4 and CH_4 have same shape
 - I_3^+ has bent geometry
 - PH_5 and BiCl_5 do not exist
 - π - π bonds are present in SO_2
15. Which of the following pairs of ions is isoelectronic and isostructural? (2016-II)
- CO_3^{2-} , NO_3^-
 - ClO_3^- , CO_3^{2-}
 - SO_3^{2-} , NO_3^-
 - ClO_3^- , SO_3^{2-}
16. Predict the correct order among the following: (2016 - I)
- Lone pair — bond pair > bond pair — bond pair > lone pair — lone pair
 - Lone pair — lone pair > lone pair — bond pair > bond pair — bond pair
 - Lone pair — lone pair > bond pair — bond pair > lone pair — bond pair
 - Bond pair — bond pair > lone pair — bond pair > lone pair — lone pair
17. Maximum bond angle at nitrogen is present in which of the following? (2015)
- NO_2^+
 - NO_3^-
 - NO_2
 - NO_2^-
18. In which of the following pairs, both the species are not isostructural? (2015)
- Diamond, Silicon carbide
 - NH_3 , PH_3
 - XeF_4 , XeO_4
 - SiCl_4 , PCl_4^+
19. Which of the following pairs of ions are isoelectronic and isostructural? (2015)
- SO_3^{2-} , NO_3^-
 - ClO_3^- , SO_3^{2-}
 - CO_3^{2-} , SO_3^{2-}
 - ClO_3^- , CO_3^{2-}
20. Which one of the following species has planar triangular shape? (2014)
- NO_3^-
 - NO_2^-
 - CO_2
 - N_3^-

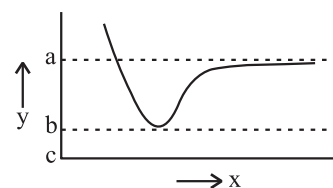
21. Which of the following is a polar molecule? (2013)
- BF_3
 - SF_4
 - SiF_4
 - XeF_4
22. XeF_2 is isostructural with (2013)
- SbCl_3
 - BaCl_2
 - TeF_2
 - ICl_2^-

Valence Bond Theory, Hybridisation

23. BF_3 is planar and electron deficient compound. Hybridization and number of electrons around the central atom, respectively are: (2021)
- sp^3 and 6
 - sp^2 and 6
 - sp^2 and 8
 - sp^3 and 4
24. How many (i) sp^2 hybridised carbon atoms and (ii) π bonds are present in the following compound? (2020-Covid)

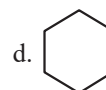
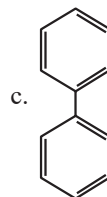
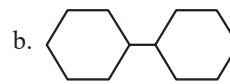
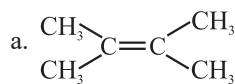


- 8, 6
 - 7, 6
 - 8, 5
 - 7, 5
25. The potential energy (y) curve for H_2 formation as a function of internuclear distance (x) of the H atoms is shown below. (2020-Covid)



The bond energy of H_2 is

- $\frac{(c-a)}{2}$
 - $\frac{(b-a)}{2}$
 - $(c-a)$
 - $(b-a)$
26. The hybridisations of atomic orbitals of nitrogen in NO_2^+ , NO_3^- and NH_4^+ respectively are: (2016-II)
- sp , sp^3 and sp^2
 - sp^2 , sp^3 and sp
 - sp , sp^2 and sp^3
 - sp^2 , sp and sp^3
27. In which of the following molecules, all atoms are coplanar? (2016 - II)



28. The correct geometry and hybridization for XeF_4 are
(2016 - II)

- Octahedral, sp^3d^2
- Trigonal bipyramidal, sp^3d
- Planar triangle, sp^3d^3
- Square planar, sp^3d^2

29. Which of the following organic compounds has same hybridisation as its combustion product (CO_2)? (2014)

- Ethyne
- Ethene
- Ethanol
- Ethane

MOT, Bonding in Some Homonuclear Diatomic Molecule

30. Which amongst the following is incorrect statement (2022)

- O_2^+ ion is diamagnetic.
- The bond orders of O_2^+ , O_2 , O_2^- and O_2^{2-} are 2.5, 2, 1.5 and 1, respectively
- C_2 molecule has four electrons in its two degenerate π molecular orbitals.
- H_2^+ ion has one electron.

31. Identify a molecule which does not exist. (2020)

- Li_2
- C_2
- O_2
- He_2

32. Which of the following diatomic molecular species has only π bonds according to Molecular Orbital Theory? (2019)

- O_2
- N_2
- C_2
- Be_2

33. Consider the following species: (2018)

CN^+ , CN^- , NO and CN

Which one of these will have the highest bond order?

- NO
- CN^-
- CN
- CN^+

34. Decreasing order of stability of O_2 , O_2^- , O_2^+ and O_2^{2-} is: (2015 Re)

- $\text{O}_2^- > \text{O}_2^{2-} > \text{O}_2^+ > \text{O}_2$
- $\text{O}_2^+ > \text{O}_2 > \text{O}_2^- > \text{O}_2^{2-}$
- $\text{O}_2^{2-} > \text{O}_2^- > \text{O}_2 > \text{O}_2^+$
- $\text{O}_2 > \text{O}_2^+ > \text{O}_2^{2-} > \text{O}_2^-$

35. Which of the following options represents the correct bond order? (2015)

- $\text{O}_2^- < \text{O}_2 < \text{O}_2^+$
- $\text{O}_2^- > \text{O}_2 < \text{O}_2^+$
- $\text{O}_2^- < \text{O}_2 > \text{O}_2^+$
- $\text{O}_2^- > \text{O}_2 > \text{O}_2^+$

36. The correct bond order in the following species is: (2015)

- $\text{O}_2^{2+} < \text{O}_2^- < \text{O}_2^+$
- $\text{O}_2^{2+} < \text{O}_2^- < \text{O}_2^+$
- $\text{O}_2^- < \text{O}_2^+ < \text{O}_2^{2+}$
- $\text{O}_2^{2+} < \text{O}_2^+ < \text{O}_2^-$

37. Which of the following is paramagnetic? (2013)

- CO
- O_2^-
- CN^-
- NO^+

Hydrogen Bonds

38. Which one of the following compounds shows the presence of intramolecular hydrogen bond? (2016 - II)

- Cellulose
- Concentrated acetic acid
- H_2O_2
- HCN

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
d	d	c	b	c	d	c	b	a	d	d	d	a	a	a,d	b	a
18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
c	b	a	b	d	b	b	d	c	c	a	a	a	d	c	b	b
35	36	37	38													
a	c	b	a													

CHAPTER 6

Coordination Compounds

Introduction and Werner's Theory of Coordination Compounds

- Ethylene diaminetetraacetate (EDTA) ion is: (2021)
 - Unidentate ligand
 - Bidentate ligand with two "N" donor atoms
 - Tridentate ligand with three "N" donor atoms
 - Hexadentate ligand with four "O" and two "N" donor atoms
- The correct order of the stoichiometries of AgCl formed when AgNO_3 in excess is treated with the complexes: $\text{CoCl}_3 \cdot 6\text{NH}_3$, $\text{CoCl}_3 \cdot 5\text{NH}_3$, $\text{CoCl}_3 \cdot 4\text{NH}_3$ respectively is: (2017-Delhi)
 - 2AgCl , 3AgCl , 1AgCl
 - 1AgCl , 3AgCl , 2AgCl
 - 3AgCl , 1AgCl , 2AgCl
 - 3AgCl , 2AgCl , 1AgCl
- Cobalt(III) chloride forms several octahedral complexes with ammonia. Which of the following will not give test for chloride ions with silver nitrate at 25°C ? (2015)
 - $\text{CoCl}_3 \cdot 4\text{NH}_3$
 - $\text{CoCl}_3 \cdot 5\text{NH}_3$
 - $\text{CoCl}_3 \cdot 6\text{NH}_3$
 - $\text{CoCl}_3 \cdot 3\text{NH}_3$
- The sum of coordination number and oxidation number of the metal M in the complex $[\text{M}(\text{en})_2(\text{C}_2\text{O}_4)]\text{Cl}$ (where en is ethylenediamine) is: (2015 Re)
 - 6
 - 7
 - 8
 - 9
- The type of isomerism shown by the complex $[\text{CoCl}_2(\text{en})_2]$ is: (2018)
 - Geometrical isomerism
 - Coordination isomerism
 - Linkage isomerism
 - Ionization isomerism
- The name of complex ion, $[\text{Fe}(\text{CN})_6]^{3-}$ is: (2015 Re)
 - Hexacyanidoferrate (III) ion
 - Hexacyanoiron (III) ion
 - Hexacyanoferrate (III) ion
 - Tricyanoferrate (III) ion
- Number of possible isomers for the complex $[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$ will be: (en = ethylenediamine) (2015 Re)
 - 4
 - 2
 - 1
 - 3

Bonding in Coordination Compounds

- The order of energy absorbed which is responsible for the color of complexes (2022)
 - $[\text{Ni}(\text{H}_2\text{O})_2(\text{en})_2]^{2+}$
 - $[\text{Ni}(\text{H}_2\text{O})_4(\text{en})]^{2+}$ and
 - $[\text{Ni}(\text{en})_3]^{2+}$
 is
 - $B > A > C$
 - $A > B > C$
 - $C > B > A$
 - $C > A > B$
- Match List-I with List-II. (2021)

List-I		List-II	
(A)	$[\text{Fe}(\text{CN})_6]^{3-}$	(i)	5.92 BM
(B)	$[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$	(ii)	0 BM
(C)	$[\text{Fe}(\text{CN})_6]^{4-}$	(iii)	4.90 BM
(D)	$[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$	(iv)	1.73 BM

Choose the correct answer from the options given below.

- The IUPAC name of the complex- (2022)
 $[\text{Ag}(\text{H}_2\text{O})_2][\text{Ag}(\text{CN})_2]$ is:
 - diaquasilver(I) dicyanidoargentate(I)
 - dicyanidosilver(II) diaquaargentate(II)
 - diaquasilver(II) dicyanidoargentate(II)
 - dicyanidosilver(I) diaquaargentate(I)

- A-ii B-iv C-iii D-i
- A-i B-iii C-iv D-ii
- A-iv B-i C-ii D-iii
- A-iv B-ii C-i D-iii

11. Which of the following is the correct order of increasing field strength of ligands to form coordination compounds? (2020)

- $\text{SCN}^- < \text{F}^- < \text{CN}^- < \text{C}_2\text{O}_4^{2-}$
- $\text{F}^- < \text{SCN}^- < \text{C}_2\text{O}_4^{2-} < \text{CN}^-$
- $\text{CN}^- < \text{C}_2\text{O}_4^{2-} < \text{SCN}^- < \text{F}^-$
- $\text{SCN}^- < \text{F}^- < \text{C}_2\text{O}_4^{2-} < \text{CN}^-$

12. Match the coordination number and type of hybridisation with distribution of hybrid orbitals in space based on Valence bond theory. (2020-Covid)

Coordination number and type of hybridisation		Distribution of hybrid orbitals in space	
(A)	4, sp^3	(i)	Trigonal bipyramidal
(B)	4, dsp^2	(ii)	Octahedral
(C)	5, sp^3d	(iii)	Tetrahedral
(D)	6, d^2sp^3	(iv)	Square planar

Select the correct option:

- A-iii B-iv C-i D-ii
- A-iv B-i C-ii D-iii
- A-iii B-i C-iv D-ii
- A-ii B-iii C-iv D-i

13. What is the correct electronic configuration of the central atom in $\text{K}_4[\text{Fe}(\text{CN})_6]$ based on crystal field theory? (2019)

- $t_{2g}^4 e_g^2$
- $t_{2g}^6 e_g^0$
- $e^3 t_{2g}^3$
- $e^4 t_{2g}^2$

14. The geometry and magnetic behaviour of the complex $[\text{Ni}(\text{CO})_4]$ are? (2018)

- Square planar geometry and diamagnetic
- Tetrahedral geometry and diamagnetic
- Tetrahedral geometry and paramagnetic
- Square planar geometry and paramagnetic

15. Correct increasing order for the wavelengths of absorption in the visible region for the complexes of Co^{3+} is: (2017-Delhi)

- $[\text{Co}(\text{NH}_3)_6]^{3+}, [\text{Co}(\text{en})_3]^{3+}, [\text{Co}(\text{H}_2\text{O})_6]^{3+}$
- $[\text{Co}(\text{en})_3]^{3+}, [\text{Co}(\text{NH}_3)_6]^{3+}, [\text{Co}(\text{H}_2\text{O})_6]^{3+}$
- $[\text{Co}(\text{H}_2\text{O})_6]^{3+}, [\text{Co}(\text{en})_3]^{3+}, [\text{Co}(\text{NH}_3)_6]^{3+}$
- $[\text{Co}(\text{H}_2\text{O})_6]^{3+}, [\text{Co}(\text{NH}_3)_6]^{3+}, [\text{Co}(\text{en})_3]^{3+}$

16. Pick out the correct statement with respect to $[\text{Mn}(\text{CN})_6]^{3-}$ (2017-Delhi)

- It is dsp^2 hybridised and square planar
- It is sp^3d^2 hybridised and octahedral
- It is sp^3d^2 hybridised and tetrahedral
- It is d^2sp^3 hybridised and octahedral

17. The correct increasing order of trans-effect of the following species is: (2016-II)

- $\text{NH}_3 > \text{CN}^- > \text{Br}^- > \text{C}_6\text{H}_5^-$
- $\text{CN}^- > \text{C}_6\text{H}_5^- > \text{Br}^- > \text{NH}_3$
- $\text{Br}^- > \text{CN}^- > \text{NH}_3 > \text{C}_6\text{H}_5^-$
- $\text{CN}^- > \text{Br}^- > \text{C}_6\text{H}_5^- > \text{NH}_3$

18. Jahn-Teller effect is not observed in high spin complexes of: [OS] (2016-II)

- d^7
- d^8
- d^4
- d^9

19. The hybridisation involved in complex $[\text{Ni}(\text{CN})_4]^{2-}$ is (Atomic Number Ni = 28) (2015 Re)

- d^2sp^3
- dsp^2
- sp^3
- d^2sp^2

20. Which of these statements about $[\text{Co}(\text{CN})_6]^{3-}$ is true? (2015)

- $[\text{Co}(\text{CN})_6]^{3-}$ has four unpaired electrons and will be in a low-spin configuration
- $[\text{Co}(\text{CN})_6]^{3-}$ has four unpaired electrons and will be in a high-spin configuration
- $[\text{Co}(\text{CN})_6]^{3-}$ has no unpaired electrons and will be in a high-spin configuration
- $[\text{Co}(\text{CN})_6]^{3-}$ has no unpaired electrons and will be in a low-spin configuration

21. Among the following complexes the one which shows zero crystal field stabilization energy (CFSE) is: (2014)

- $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$
- $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$
- $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$
- $[\text{Mn}(\text{H}_2\text{O})_6]^{3+}$

22. A magnetic moment of 1.73 BM will be shown by one among the following: (2013)

- $[\text{Cu}(\text{NH}_3)_4]^{2+}$
- $[\text{Ni}(\text{CN})_4]^{2-}$
- TiCl_4
- $[\text{CoCl}_6]^{4-}$

Bonding in Metal Carbonyls

23. Iron carbonyl, $\text{Fe}(\text{CO})_5$ is (2018)

- Tetranuclear
- Mononuclear
- Dinuclear
- Trinuclear

24. Which of the following has longest C-O bond length? (Free C—O bond length in CO is 1.128 Å) (2016 - I)

- $[\text{Mn}(\text{CO})_6]^+$
- $\text{Ni}(\text{CO})_4$
- $[\text{Co}(\text{CO})_4]^+$
- $[\text{Fe}(\text{CO})_4]^{2-}$

Applications of Coordination Compounds

25. Which of the following complexes is used to be as an anticancer agent? (2014)

- cis- $[\text{PtCl}_2(\text{NH}_3)_2]$
- cis- $\text{K}_2[\text{PtCl}_2\text{Br}_2]$
- $\text{Na}_2[\text{CoCl}_4]$
- mer- $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
d	d	d	d	a	a	a	d	d	c	d	a	b	b	b	d	b
18	19	20	21	22	23	24	25									
b	b	d	a	a	b	d	a									

CHAPTER 7

Some Basic Concepts of Chemistry

Significant Figures, Laws of Chemical Combinations

1. Equal masses of H_2 , O_2 and methane have been taken in a container of volume V at temperature $27^\circ C$ in identical conditions. The ratio of the volumes of gases $H_2 : O_2$: methane would be: (2014)
- a. 8 : 16 : 1 b. 16 : 8 : 1
c. 16 : 1 : 2 d. 8 : 1 : 2

Atomic and Molecular Masses

2. Suppose the elements X and Y combine to form two compounds XY_2 and X_3Y_2 . When 0.1 mole of XY_2 weighs 10 g and 0.05 mole of X_3Y_2 weighs 9 g, the atomic weights of X and Y are: (2016)
- a. 20, 30 b. 30, 20
c. 40, 30 d. 60, 40

Mole Concept and Molar Masses

3. Which one of the followings has maximum number of atoms? (2020)
- a. 1 g of Mg(s) [Atomic mass of Mg = 24]
b. 1 g of O_2 (g) [Atomic mass of O = 16]
c. 1 g of Li(s) [Atomic mass of Li = 7]
d. 1 g of Ag(s) [Atomic mass of Ag = 108]
4. One mole of carbon atom weighs 12g, the number of atoms in it is equal to. (2020-Covid)
- (Mass of carbon- 12 is 1.9926×10^{-23} g)
- a. 6.022×10^{22} b. 12×10^{22}
c. 6.022×10^{23} d. 12×10^{23}
5. In which case is number of molecules of water maximum? (2018)
- a. 18 mL of water
b. 0.18 g of water
c. 10^{-3} mol of water
d. 0.00224 L of water vapours at 1 atm and 273 K

6. A mixture of gases contains H_2 and O_2 gases in the ratio of 1 : 4 (w/w). What is the molar ratio of the two gases in the mixture? (2015)
- a. 16 : 1 b. 2 : 1
c. 1 : 4 d. 4 : 1
7. The number of water molecules is maximum in: (2015 Re)
- a. 18 moles of water b. 18 molecules of water
c. 1.8 gram of water d. 18 gram of water
8. If Avogadro number N_A , is changed from $6.022 \times 10^{23} \text{ mol}^{-1}$ to $6.022 \times 10^{20} \text{ mol}^{-1}$, this would change: (2015 Re)
- a. The ratio of elements to each other in a compound
b. The definition of mass in units of grams
c. The mass of one mole of carbon
d. The ratio of chemical species to each other in a balanced equation

Percentage Composition, Empirical & Molecular Formula

9. An organic compound contains 78% (by wt.) carbon and remaining percentage of hydrogen. The right option for the empirical formula of this compound is: [Atomic wt. of C is 12, H is 1] (2021)
- a. CH_2 b. CH_3
c. CH_4 d. CH

Stoichiometry and Stoichiometric Calculations

10. The number of moles of hydrogen molecules required to produce 20 moles of ammonia through Haber's process is: (2019)
- a. 10 b. 20
c. 30 d. 40
11. A mixture of 2.3 g formic acid and 4.5 g oxalic acid is treated with concentration H_2SO_4 . The evolved gaseous mixture is passed through KOH pellets. Weight (in g) of the remaining product at STP will be: (2018)
- a. 1.4 b. 3.0
c. 4.4 d. 2.8

12. What is the mass of the precipitate formed when 50 mL of 16.9% solution of AgNO_3 is mixed with 50 mL of 5.8% NaCl solution?

($\text{Ag} = 107.8$, $\text{N} = 14$, $\text{O} = 16$, $\text{Na} = 23$, $\text{Cl} = 35.5$) (2015 Re)

- a. 3.5 g b. 7 g
c. 14 g d. 28 g

13. 20.0 g of a magnesium carbonate sample decomposes on heating to give carbon dioxide and 8.0 g magnesium oxide. What will be the percentage purity of magnesium carbonate in the sample? (2015 Re)

(Atomic weight of $\text{Mg} = 24$)

- a. 96 b. 60
c. 84 d. 75

14. When 22.4 litres of $\text{H}_2(\text{g})$ is mixed with 11.2 litres of $\text{Cl}_2(\text{g})$, each at STP, the moles of $\text{HCl}(\text{g})$ formed is equal to: (2014)

- a. 2 mol of $\text{HCl}(\text{g})$ b. 0.5 mol of $\text{HCl}(\text{g})$
c. 1.5 mol of $\text{HCl}(\text{g})$ d. 1 mol of $\text{HCl}(\text{g})$

15. 1.0 g of magnesium is burnt with 0.56 g O_2 in a closed vessel. Which reactant is left in excess and how much? (2014)

(Atomic weight $\text{Mg} = 24$; $\text{O} = 16$)

- a. O_2 , 0.16 g b. Mg , 0.44 g
c. O_2 , 0.28 g d. Mg , 0.16 g

Concentration Terms

16. What mass of 95% pure CaCO_3 will be required to neutralise 50 mL of 0.5 M HCl solution according to the following reaction? (2022)



[Calculate upto second place of decimal point]

- a. 9.50 g b. 1.25 g
c. 1.32 g d. 3.66 g

17. 6.02×10^{20} molecules of urea are present in 100 mL of its solution. The concentration of solution is: (2013)

- a. 0.02 M b. 0.01 M
c. 0.001 M d. 0.1 M

18. An excess of AgNO_3 is added to 100 mL of a 0.01 M solution of dichlorotetraaquachromium(III) chloride. The number of moles of AgCl precipitated would be: (2013)

- a. 0.001 b. 0.002
c. 0.003 d. 0.01

19. How many grams of concentrated nitric acid solution should be used to prepare 250 mL of 2.0 M HNO_3 ? The concentrated acid is 70 % HNO_3 : (2013)

- a. 45.0 g conc. HNO_3 b. 90.0 g conc. HNO_3
c. 70.0 g conc. HNO_3 d. 54.0 g conc. HNO_3

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
c	c	c	c	a	d	a	c	b	c	d	b	c	d	d	c	b
18	19															
a	a															

CHAPTER 8

Solutions

Expressing Concentration of Solutions

- In one molal solution that contains 0.5 mole of a solute, there is (2022)
 - 1000 g of solvent
 - 500 mL of solvent
 - 500 g of solvent
 - 100 mL of solvent
- Which of the following is dependent on temperature? (2017-Delhi)
 - Weight percentage
 - Molality
 - Molarity
 - Mole fraction
- What is the mole fraction of the solute in a 1.00 m aqueous solution? (2015 Re)
 - 0.0177
 - 0.177
 - 1.770
 - 0.0354
- How many grams of concentrated nitric acid solution should be used to prepare 250 mL of 2.0 M HNO_3 ? The concentrated acid is 70% HNO_3 . (2013)
 - 70.0 g conc. HNO_3
 - 54.0 g conc. HNO_3
 - 45.0 g conc. HNO_3
 - 90.0 g conc. HNO_3
- The correct option for the value of vapour pressure of a solution at 45°C with benzene to octane in molar ratio 3 : 2 is: (2021)

[At 45°C vapour pressure of benzene is 280 mm Hg and that of octane is 420 mm Hg. Assume Ideal gas]

 - 168 mm of Hg
 - 336 mm of Hg
 - 350 mm of Hg
 - 160 mm of Hg
- Which of the following statements about the composition of the vapour over an ideal 1 : 1 molar mixture of benzene and toluene is correct? Assume that the temperature is at 25°C. (Given, vapour pressure data at 25°C, benzene = 12.8 kPa, toluene = 3.85 kPa) (2016-I)
 - The vapour will contain equal amounts of benzene and toluene
 - Not enough information is given to make a prediction
 - The vapour will contain a higher percentage of benzene
 - The vapour will contain a higher percentage of toluene

Ideal and Non-Ideal Solutions

Vapour Pressure of Liquid Solutions

- Which one is not correct mathematical equation for Dalton's Law of partial pressure? Here p = total pressure of gaseous mixture (2022)
 - $p_i = x_i p_i^\circ$, where x_i = mole fraction of i^{th} gas in gaseous mixture
 p_i° = pressure of i^{th} gas in pure state
 - $p = p_1 + p_2 + p_3$
 - $p = n_1 \frac{RT}{V} + n_2 \frac{RT}{V} + n_3 \frac{RT}{V}$
 - $p_i = x_i p$, where p_i = partial pressure of i^{th} gas
 x_i = mole fraction of i^{th} gas in gaseous mixture
- The mixture which shows positive deviation from Raoult's law is: (2020)
 - Benzene + Toluene
 - Acetone + Chloroform
 - Chloroethane + Bromoethane
 - Ethanol + Acetone
- For an ideal solution, the correct option is : (2019)
 - $\Delta_{\text{mix}} S = 0$ at constant T and P
 - $\Delta_{\text{mix}} V \neq 0$ at constant T and P
 - $\Delta_{\text{mix}} H = 0$ at constant T and P
 - $\Delta_{\text{mix}} G = 0$ at constant T and P
- The mixture that forms maximum boiling azeotrope is: (2019)
 - Water + Nitric acid
 - Ethanol + Water
 - Acetone + Carbon disulphide
 - Heptane + Octane

11. Which one of the following is incorrect for ideal solution? (2016 - II)

- a. $\Delta P = P_{\text{obs}} - P_{\text{calculated by Raoult's law}} = 0$
- b. $\Delta G_{\text{mix}} = 0$
- c. $\Delta H_{\text{mix}} = 0$
- d. $\Delta U_{\text{mix}} = 0$

12. Which one is not equal to zero for an ideal solution? (2015)

- a. ΔS_{mix}
- b. ΔV_{mix}
- c. $\Delta P = P_{\text{observed}} - P_{\text{Raoult}}$
- d. ΔH_{mix}

Colligative Properties and Determination of Molar Mass

13. The following solutions were prepared by dissolving 10 g of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) in 250 ml of water (P_1), 10 g of urea ($\text{CH}_4\text{N}_2\text{O}$) in 250 ml of water (P_2) and 10 g of sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) in 250 ml of water (P_3). The right option for the decreasing order of osmotic pressure of these solutions is: (2021)

- a. $P_1 > P_2 > P_3$
- b. $P_2 > P_3 > P_1$
- c. $P_3 > P_1 > P_2$
- d. $P_2 > P_1 > P_3$

14. If 8 g of a non-electrolyte solute is dissolved in 114 g of n-octane to reduce its vapour pressure to 80%, the molar mass (in g mol^{-1}) of the solute is [Given that molar mass of n-octane is 114 g mol^{-1}] (2020-Covid)

- a. 60
- b. 80
- c. 20
- d. 40

15. Isotonic solutions have same (2020-Covid)

- a. Freezing temperature
- b. Osmotic pressure
- c. Boiling temperature
- d. Vapour pressure

16. If molality of the dilute solution is doubled, the value of molal depression constant (K_f) will be: (2017-Delhi)

- a. Unchanged
- b. Doubled
- c. Halved
- d. Tripled

17. At 100°C , the vapour pressure of a solution of 6.5 g of a solute in 100 g water is 732 mm. If $K_b = 0.52$, the boiling point of this solution will be: (2016 - I)

- a. 103°C
- b. 101°C
- c. 100°C
- d. 102°C

Abnormal Molar Mass

18. The freezing point depression constant (K_f) of benzene is $5.12 \text{ K kg mol}^{-1}$. The freezing point depression for the solution of molality 0.078 m containing a non-electrolyte solute in benzene is (rounded off upto two decimal places): (2020)

- a. 0.80 K
- b. 0.40 K
- c. 0.60 K
- d. 0.20 K

19. The van't Hoff factor (i) for a dilute aqueous solution of the strong electrolyte barium hydroxide is: (2016 - II)

- a. 2
- b. 3
- c. 0
- d. 1

20. Which one of the following electrolytes has the same value of van't Hoff's factor (i) as that of $\text{Al}_2(\text{SO}_4)_3$ (if all are 100% ionised)? (2015)

- a. $\text{K}_3[\text{Fe}(\text{CN})_6]$
- b. $\text{Al}(\text{NO}_3)_3$
- c. $\text{K}_4[\text{Fe}(\text{CN})_6]$
- d. K_2SO_4

21. The boiling point of 0.2 mol kg^{-1} solution of X in water is greater than equimolal solution of Y in water. Which one of the following statements is true in this case? (2015)

- a. Molecular mass of X is greater than the molecular mass of Y
- b. Molecular mass of X is less than the molecular mass of Y
- c. Y is undergoing dissociation in water while X undergoes no change
- d. X is undergoing dissociation in water

22. Of the following 0.10 m aqueous solutions, which one will exhibit the largest freezing point depression? (2014)

- a. $\text{C}_6\text{H}_{12}\text{O}_6$
- b. $\text{Al}_2(\text{SO}_4)_3$
- c. K_2SO_4
- d. KCl

Answer Key

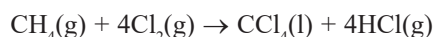
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
c	c	a	c	a	b	c	d	c	a	b	a	d	d	b	a	b
18	19	20	21	22												
b	b	c	d	b												

CHAPTER 9

Redox Reactions

Redox Reactions (Oxidation and Reduction), Oxidation Number

1. What is the change in oxidation number of carbon in the following reaction? (2020)



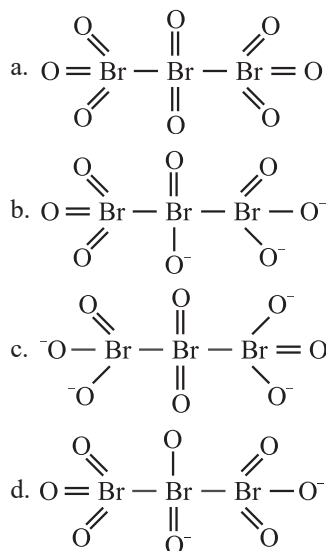
- a. 0 to +4 b. -4 to +4
c. 0 to -4 d. +4 to +4

2. The oxidation number of the underlined atom in the following species (2020-Covid)

- a. $\text{Cl}\underline{\text{O}}_3^-$ is +5 b. $\text{K}_2\underline{\text{Cr}}_2\text{O}_7$ is +6
c. $\text{H}\underline{\text{Au}}\text{Cl}_4$ is +3 d. $\text{Cu}_2\underline{\text{O}}$ is -1

Identify the incorrect option

3. The correct structure of tribromooctaoxide is (2019)



4. Hot concentrated sulphuric acid is a moderately strong oxidising agent. Which of the following reactions does not show oxidising behaviour? (2016 - I)

- a. $\text{C} + 2\text{H}_2\text{SO}_4 \rightarrow \text{CO}_2 + 2\text{SO}_2 + 2\text{H}_2\text{O}$
b. $\text{CaF}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + 2\text{HF}$
c. $\text{Cu} + 2\text{H}_2\text{SO}_4 \rightarrow \text{CuSO}_4 + \text{SO}_2 + 2\text{H}_2\text{O}$
d. $3\text{S} + 2\text{H}_2\text{SO}_4 \rightarrow 3\text{SO}_2 + 2\text{H}_2\text{O}$

5. In acidic medium, H_2O_2 changes $\text{Cr}_2\text{O}_7^{2-}$ to CrO_5 which has two ($-\text{O}-\text{O}-$) bonds. Oxidation state of Cr in CrO_5 is: (2014)

- a. +3 b. +6
c. -10 d. +5

6. The pair of compounds that can exist together is (2014)

- a. FeCl_3 , SnCl_2
b. HgCl_2 , SnCl_2
c. FeCl_2 , SnCl_2
d. FeCl_3 , KI

7. The oxidation state of Cr in CrO_5 is (2014)

- a. -6 b. +12
c. +6 d. +4

Types of Redox Reactions and Balancing of Redox Reactions

8. Which of the following reactions is the metal displacement reaction? Choose the right option. (2021)

- a. $\text{Cr}_2\text{O}_3 + 2\text{Al} \xrightarrow{\Delta} \text{Al}_2\text{O}_3 + 2\text{Cr}$
b. $\text{Fe} + 2\text{HCl} \longrightarrow \text{FeCl}_2 + \text{H}_2 \uparrow$
c. $2\text{Pb}(\text{NO}_3)_2 \longrightarrow 2\text{PbO} + 4\text{NO}_2 + \text{O}_2 \uparrow$
d. $2\text{KClO}_3 \xrightarrow{\Delta} 2\text{KCl} + 3\text{O}_2$

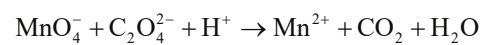
9. Which of the following reactions are disproportionation reaction? (2019)

- A. $2\text{Cu}^+ \longrightarrow \text{Cu}^{2+} + \text{Cu}^0$
B. $3\text{MnO}_4^{2-} + 4\text{H}^+ \longrightarrow 2\text{MnO}_4^- + \text{MnO}_2 + 2\text{H}_2\text{O}$
C. $2\text{KMnO}_4 \xrightarrow{\Delta} \text{K}_2\text{MnO}_4 + \text{MnO}_2 + \text{O}_2$
D. $2\text{MnO}_4^- + 3\text{Mn}^{2+} + 2\text{H}^+ \longrightarrow 5\text{MnO} + 4\text{H}^+$

Select the correct option from the following

- a. (A) and (B) only
b. (A), (B) and (C)
c. (A), (C) and (D)
d. (A) and (D) only

10. For the redox reaction



The correct coefficients of the reactants for the balanced equation are:
(2018)

	MnO_4^-	$\text{C}_2\text{O}_4^{2-}$	H^+
a.	16	5	2
b.	2	5	16
c.	5	16	2
d.	2	16	5

Answer Key

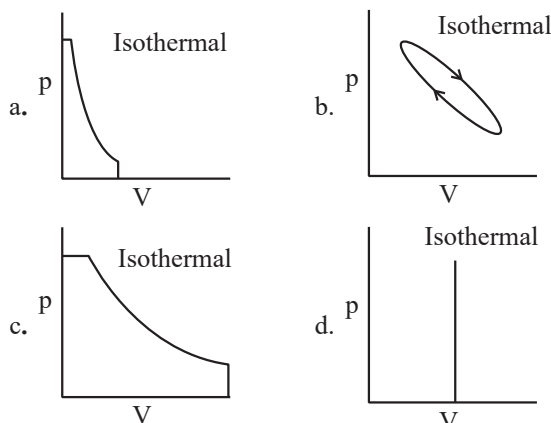
1	2	3	4	5	6	7	8	9	10
b	d	a	b	b	c	c	a	a	b

CHAPTER 10

Thermodynamics

Measurement of ΔU & ΔH , Work Done and Heat Capacity

1. Which of the following p-V curve represents maximum work done? (2022)



2. Which one among the following is the correct option for right relationship between C_p and C_v for one mole of ideal gas? (2021)

- a. $C_p - C_v = R$ b. $C_p = RC_v$
c. $C_v = RC_p$ d. $C_p + C_v = R$

3. The correct option for free expansion of an ideal gas under adiabatic condition is : (2020)

- a. $q = 0$, $\Delta T < 0$ and $w > 0$ b. $q < 0$, $\Delta T = 0$ and $w = 0$
c. $q > 0$, $\Delta T > 0$ and $w > 0$ d. $q = 0$, $\Delta T = 0$ and $w = 0$

4. Under isothermal condition, a gas at 300 K expands from 0.1 L to 0.25 L against a constant external pressure of 2 bar. The work done by the gas is (Given that 1 L bar = 100 J) (2019)

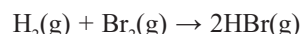
- a. -30 J b. 5 kJ
c. 25 J d. 30 J

5. A gas is allowed to expand in a well insulated container against a constant external pressure of 2.5 atm from an initial volume of 2.50 L to a final volume of 4.50 L. The change in internal energy ΔU of the gas in joules will be: (2017-Delhi)

- a. +505 J b. 1136.25 J
c. -505 J d. -505 J

Enthalpies For Different Types of Reactions

6. At standard conditions, if the change in the enthalpy for the following reaction is -109 kJ mol^{-1}



Given that bond energy of H_2 and Br_2 is 435 kJ mol^{-1} and 192 kJ mol^{-1} , respectively, what is the bond energy (in kJ mol^{-1}) of HBr? (2020-Covid)

- a. 736 b. 518
c. 259 d. 368

7. The bond dissociation energies of X_2 , Y_2 and XY are in the ratio of 1 : 0.5 : 1. ΔH for the formation of XY is -200 kJ mol^{-1} (2018)

The bond dissociation energy of X_2 will be

- a. 200 kJ mol^{-1} b. 100 kJ mol^{-1}
c. 400 kJ mol^{-1} d. 800 kJ mol^{-1}

8. The heat of combustion of carbon to CO_2 is -393.5 kJ/mol . The heat released upon formation of 35.2 g of CO_2 from carbon and oxygen gas is: (2015)

- a. + 315 kJ b. - 630 kJ
c. - 3.15 kJ d. - 315 kJ

Spontaneity, Entropy, Gibbs Energy Change and Equilibrium

9. For irreversible expansion of an ideal gas under isothermal condition, the correct option is: (2021)

- a. $\Delta U \neq 0$, $\Delta S_{\text{total}} \neq 0$ b. $\Delta U = 0$, $\Delta S_{\text{total}} \neq 0$
c. $\Delta U \neq 0$, $\Delta S_{\text{total}} = 0$ d. $\Delta U = 0$, $\Delta S_{\text{total}} = 0$

10. For the reaction, $2\text{Cl}(\text{g}) \rightarrow \text{Cl}_2(\text{g})$, the correct option is: (2020)

- a. $\Delta_r H > 0$ and $\Delta_r S < 0$
b. $\Delta_r H < 0$ and $\Delta_r S < 0$
c. $\Delta_r H < 0$ and $\Delta_r S < 0$
d. $\Delta_r H > 0$ and $\Delta_r S > 0$

11. If for a certain reaction $\Delta_r H$ is 30 kJ mol^{-1} at 450 K , the value of $\Delta_r S$ (in $\text{JK}^{-1} \text{ mol}^{-1}$) for which the same reaction will be spontaneous at the same temperature is (2020-Covid)
 a. -33 b. 33
 c. -70 d. 70
12. In which case change in entropy is negative? (2019)
 a. Evaporation of water
 b. Expansion of a gas at constant temperature
 c. Sublimation of solid to gas
 d. $2\text{H(g)} \rightarrow \text{H}_2\text{(g)}$
13. For a given reaction, $\Delta H = 35.5 \text{ KJ mol}^{-1}$ and $\Delta S = 83.6 \text{ JK}^{-1} \text{ mol}^{-1}$. The reaction is spontaneous at: (Assume that ΔH and ΔS do not vary with temperature) (2017-Delhi)
 a. $T > 298 \text{ K}$ b. $T < 425 \text{ K}$
 c. $T > 425 \text{ K}$ d. All temperatures
14. For a sample of perfect gas when its pressure is changed isothermally from P_i to P_f , the entropy change is given by: (2016 - II)
 a. $\Delta S = nRT \ln \left(\frac{P_f}{P_i} \right)$ b. $\Delta S = nRT \ln \left(\frac{P_i}{P_f} \right)$
 c. $\Delta S = nR \ln \left(\frac{P_f}{P_i} \right)$ d. $\Delta S = nR \ln \left(\frac{P_i}{P_f} \right)$
15. The correct thermodynamic conditions for the spontaneous reaction at all temperatures is: (2016 - I)
 a. $\Delta H < 0$ and $\Delta S < 0$
 b. $\Delta H < 0$ and $\Delta S = 0$
 c. $\Delta H > 0$ and $\Delta S < 0$
 d. $\Delta H < 0$ and $\Delta S > 0$
16. For the reaction, $\text{X}_2\text{O}_4(\text{l}) \rightarrow 2\text{XO}_2(\text{g})$, $\Delta U = 2.1 \text{ kcal}$, $\Delta S = 20 \text{ cal K}^{-1}$ at 300 K . Hence, ΔG is: (2014)
 a. -2.7 kcal b. 9.3 kcal
 c. -9.3 kcal d. 2.7 kcal

Clausius Clapeyron Equation

17. Consider the following liquid-vapour equilibrium.

Liquid $\rightleftharpoons$ Vapour. Which of the following relations is correct? (2016 - I)

- a. $\frac{d \ln P}{dT} = \frac{-\Delta H_v}{T^2}$ b. $\frac{d \ln P}{dT} = \frac{-\Delta H_v}{RT^2}$
 c. $\frac{d \ln G}{dT} = \frac{\Delta H_v}{RT^2}$ d. $\frac{d \ln P}{dT} = \frac{-\Delta H_v}{RT}$

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
c	a	d	a	d	d	d	a	b	c	d	d	c	d	d	a	b

CHAPTER 11

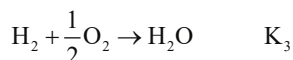
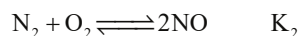
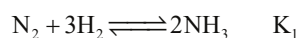
Equilibrium

Law of Chemical Equilibrium, Equilibrium Const. and Its Application

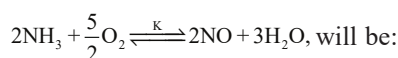
1. $3\text{O}_2(\text{g}) \rightleftharpoons 2\text{O}_3(\text{g})$ for the given reaction at 298 K, K_c is found to be 3.0×10^{-59} . If the concentration of O_2 at equilibrium is 0.040 M, then concentration of O_3 in M is: (2022)

- a. 1.2×10^{21} b. 4.38×10^{-32}
c. 1.9×10^{-63} d. 2.4×10^{31}

2. The equilibrium constants of the following are:

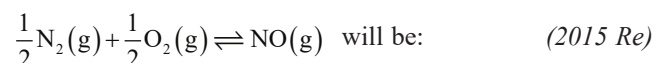


The equilibrium constant (K) of the reaction: (2017-Delhi)



- a. $K_2^3 K_3 / K_1$ b. $K_1 K_3^3 / K_2$
c. $K_1 K_3^3 / K$ d. $K_2 K_3 / K_1$

3. If the equilibrium constant for



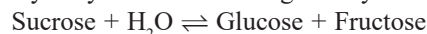
- a. K^2 b. $K^{1/2}$
c. $\frac{1}{2}K$ d. K

4. If the value of an equilibrium constant for a particular reaction is 1.6×10^{12} , then at equilibrium the system will contain: (2015)

- a. Mostly reactants
b. Mostly products
c. Similar amounts of reactants and products
d. All reactants

Relation Between K, Q and G, Factors Affecting Equilibria

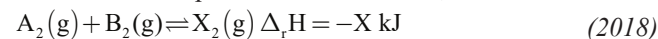
5. Hydrolysis of sucrose is given by the following reaction:



If the equilibrium constant (K_c) is 2×10^{13} at 300 K, the value of $\Delta_r G^\circ$ at the same temperature will be : (2020)

- a. $8.314 \text{ J mol}^{-1} \text{ K}^{-1} \times 300 \text{ K} \times \ln (2 \times 10^{13})$
b. $8.314 \text{ J mol}^{-1} \text{ K}^{-1} \times 300 \text{ K} \times \ln (3 \times 10^{13})$
c. $-8.314 \text{ J mol}^{-1} \text{ K}^{-1} \times 300 \text{ K} \times \ln (4 \times 10^{13})$
d. $-8.314 \text{ J mol}^{-1} \text{ K}^{-1} \times 300 \text{ K} \times \ln (2 \times 10^{13})$

6. Which one of the following conditions will favour maximum formation of the product in the reaction,



- a. Low temperature and high pressure
b. Low temperature and low pressure
c. High temperature and low pressure
d. High temperature and high pressure

7. Which one of the following statements is not correct? (2017-Delhi)

- a. Coenzymes increase the catalytic activity of enzyme
b. Catalyst does not initiate any reaction
c. The value of equilibrium constant is changed in the presence of a catalyst in the reaction at equilibrium
d. Enzymes catalyse mainly bio-chemical reactions

8. A 20 litre container at 400 K contains $\text{CO}_2(\text{g})$ at pressure 0.4 atm and an excess of SrO (neglect the volume of solid SrO). The volume of the containers is now decreased by moving the movable piston fitted in the container. The maximum volume of the container, when pressure of CO_2 attains its maximum value, will be: (2017-Delhi)



- a. 2 litre b. 5 litre
c. 10 litre d. 4 litre

9. Consider the nitration of benzene using mixed conc. H_2SO_4 and HNO_3 . If a larger amount of KHSO_4 is added to the mixture, the rate of nitration will be: (2016-I)

- a. Doubled b. Increase
c. Decrease d. Unchanged

10. Which of the following statements is correct for a reversible process in a state of equilibrium? (2015)

- a. $\Delta G^\circ = -2.303 RT \log K$ b. $\Delta G^\circ = 2.303 RT \log K$
c. $\Delta G = -2.303 RT \log K$ d. $\Delta G = 2.303 RT \log K$

11. For a given exothermic reaction, K_p and K'_p are the equilibrium constants at temperature T_1 and T_2 , respectively. Assuming that heat of reaction is constant in temperature range between T_1 and T_2 , it is readily observed that (2014)

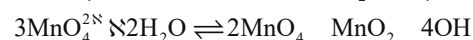
- a. $K_p > K'_p$ b. $K_p < K'_p$
c. $K_p = K'_p$ d. $K_p = \frac{1}{K'_p}$

12. For the reversible reaction,

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) + \text{Heat}$. The equilibrium shifts in forward direction: (2014)

- a. By decreasing the pressure
b. By decreasing the concentrations of $N_2(g)$ and $H_2(g)$
c. By increasing pressure and decreasing temperature
d. By increasing the concentration of $NH_3(g)$

13. $KMnO_4$ can be prepared from K_2MnO_4 as per the reaction:



The reaction can go to completion by removing OH^- ions by adding: (2013)

- a. HCl b. KOH
c. CO_2 d. SO_2

Acids, Bases and Salts

14. Conjugate base for Bronsted acids H_2O and HF are: (2019)

- a. OH^- and H_2F^+ , respectively
b. H_3O^+ and F^- , respectively
c. OH^- and F^- , respectively
d. H_3O^+ and H_2F^+ , respectively

15. Which of the following fluoro-compounds is most likely to behave as a Lewis base? (2016-II)

- a. CF_4 b. SiF_4
c. BF_3 d. PF_3

16. Which of these is least likely to act as a Lewis base? (2013)

- a. CO b. F^-
c. BF_3 d. PF_3

Ionization of Acids & Bases, pH Scale, Hydrolysis

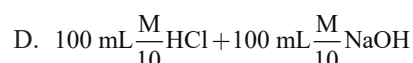
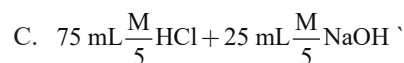
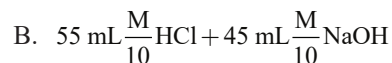
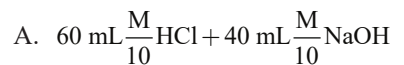
17. The pK_b of dimethylamine and pK_a of acetic acid are 3.27 and 4.77 respectively at $T(K)$. The correct option for the pH of dimethylammonium acetate solution is: (2021)

- a. 5.50 b. 7.75
c. 6.25 d. 8.50

18. Which among the following salt solutions is basic in nature? (2020-Covid)

- a. Ammonium sulphate b. Ammonium nitrate
c. Sodium acetate d. Ammonium chloride

19. Following solutions were prepared by mixing different volumes of NaOH and HCl of different concentrations: (2018)



pH of which one of them will be equal to 1?

- a. B b. A
c. C d. D

20. The percentage of pyridine (C_5H_5N) that forms pyrimidine ion ($C_5H_5N^+H$) in a 0.10 M aqueous pyridine solution (K_b for $C_5H_5N = 1.7 \times 10^{-9}$) is: (2016-II)

- a. 0.77% b. 1.6%
c. 0.0060% d. 0.013%

21. What is the pH of the resulting solution when equal volumes of 0.1 M NaOH and 0.01 M HCl are mixed? (2015 Re)

- a. 2.0 b. 7.0
c. 1.04 d. 12.65

22. Aqueous solution of which of the following compounds is the best conductor of electric current? (2015 Re)

- a. Hydrochloric acid, HCl b. Ammonia, NH_3
c. Fructose, $C_6H_{12}O_6$ d. Acetic acid, $C_2H_4O_2$

23. Which of the following salts will give highest pH in water? (2014)

- a. NaCl b. Na_2CO_3
c. $CuSO_4$ d. KCl

Buffer Solutions

24. The pH of the solution containing 50 mL each of 0.10 M sodium acetate and 0.01 M acetic acid is (2022)

[Given pK_a of $CH_3COOH = 4.57$]

- a. 2.57 b. 5.57
c. 3.57 d. 4.57

25. Which will make basic buffer? (2019)

- a. 50 mL of 0.1 M NaOH + 25 mL of 0.1 M CH_3COOH
b. 100 mL of 0.1 M CH_3COOH + 100 mL of 0.1 M NaOH
c. 100 mL of 0.1 M HCl + 200 mL of 0.1 M NH_4OH
d. 100 mL of 0.1 M HCl + 100 mL of 0.1 M NaOH

26. Which one of the following pairs of solution is not an acidic buffer? (2015 Re)

- a. H_3PO_4 and Na_3PO_4
b. $HClO_4$ and $NaClO_4$
c. CH_3COOH and CH_3COONa
d. H_2CO_3 and Na_2CO_3

Solubility Equilibria of Sparingly Soluble Salts

27. Find out the solubility of $\text{Ni}(\text{OH})_2$ in 0.1 M NaOH. Given that the ionic product of $\text{Ni}(\text{OH})_2$ is 2×10^{-15} . (2020)
- a. 2×10^{-8} M b. 1×10^{-13} M
c. 1×10^8 M d. 2×10^{-13} M
28. HCl was passed through a solution of CaCl_2 , MgCl_2 and NaCl. Which of the following compounds crystallises? (2020)
- a. Only NaCl b. Only MgCl_2
c. NaCl, MgCl_2 and CaCl_2 d. Both MgCl_2 and CaCl_2
29. The solubility product for a salt of the type AB is 4×10^{-8} . What is the molarity of its standard solution? (2020-Covid)
- a. 16×10^{-16} mol/L b. 2×10^{-16} mol/L
c. 4×10^{-4} mol/L d. 2×10^{-4} mol/L
30. pH of a saturated solution of $\text{Ca}(\text{OH})_2$ is 9. The solubility product (K_{sp}) of $\text{Ca}(\text{OH})_2$ is: (2019)
- a. 0.5×10^{-15} b. 0.25×10^{-10}
c. 0.125×10^{-15} d. 0.5×10^{-10}
31. The solubility of BaSO_4 in water is 2.42×10^{-3} g L⁻¹ at 298 K. The value of its solubility product (K_{sp}) will be (2018)
- (Given molar mass of $\text{BaSO}_4 = 233$ g mol⁻¹)
- a. 1.08×10^{-10} mol² L⁻² b. 1.08×10^{-12} mol² L⁻²
c. 1.08×10^{-8} mol² L⁻² d. 1.08×10^{-14} mol² L⁻²
32. Concentration of the Ag^+ ions in a saturated solution of $\text{Ag}_2\text{C}_2\text{O}_4$ is 2.2×10^{-4} mol L⁻¹. Solubility product of $\text{Ag}_2\text{C}_2\text{O}_4$ is: (2017-Delhi)
- a. 5.3×10^{-12} b. 2.42×10^{-8}
c. 2.66×10^{-12} d. 4.5×10^{-11}
33. The solubility of $\text{AgCl}(\text{s})$ with solubility product 1.6×10^{-10} in 0.1 M NaCl solution would be: (2016-II)
- a. 1.6×10^{-11} M b. Zero
c. 1.26×10^{-5} M d. 1.6×10^{-9} M
34. MY and NY_3 , two nearly insoluble salts, have the same K_{sp} values of 6.2×10^{-13} at room temperature, which statements would be true in regard to MY and NY_3 ? (2016-I)
- a. The addition of the salt of KY to solution of MY and NY_3 will have no effect on their solubilities
b. The molar solubilities of MY and NY_3 in water are identical
c. The molar solubility of MY in water is less than that of NY_3
d. The salts MY and NY_3 are more soluble in 0.5 M KY than in pure water
35. The K_{sp} of Ag_2CrO_4 , AgCl , AgBr and AgI are respectively, 1.1×10^{-12} , 1.8×10^{-10} , 5.0×10^{-13} , 8.3×10^{-17} . Which one of the following salts will precipitate last if AgNO_3 solution is added to the solution containing equal moles of NaCl, NaBr, NaI and Na_2CrO_4 ? (2015)
- a. AgCl b. AgBr
c. Ag_2CrO_4 d. AgI
36. Using the Gibb's energy change, $\Delta G^\circ = + 63.3$ kJ, for the following reaction,
- $$\text{Ag}_2\text{CO}_3(\text{s}) \rightleftharpoons 2\text{Ag}^+(\text{aq}) + \text{CO}_3^{2-}(\text{aq})$$
- the K_{sp} of $\text{Ag}_2\text{CO}_3(\text{s})$ in water at 25°C is ($R = 8.314$ J K⁻¹ mol⁻¹) (2014)
- a. 3.2×10^{-26} b. 8.0×10^{-12}
c. 2.9×10^{-3} d. 7.9×10^{-2}
37. Identify the correct order of solubility in aqueous medium: (2013)
- a. $\text{Na}_2\text{S} > \text{CuS} > \text{ZnS}$
b. $\text{Na}_2\text{S} > \text{ZnS} > \text{CuS}$
c. $\text{CuS} > \text{ZnS} > \text{Na}_2\text{S}$
d. $\text{ZnS} > \text{Na}_2\text{S} > \text{CuS}$

Answer Key

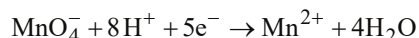
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b	c	b	b	d	a	c	b	c	a	a	c	c	c	d	c	b
18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
c	c	d	d	a	b	b	c	b	d	a	d	a	a	a	d	c
35	36	37														
c	b	b														

CHAPTER 12

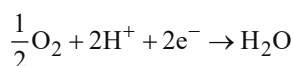
Electrochemistry

Electrochemical and Galvanic Cells

1. Given below are half cell reactions: (2022)



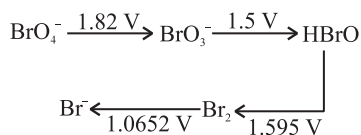
$$E^\circ_{\text{Mn}^{2+}/\text{MnO}_4^-} = -1.510 \text{ V}$$



$$E_{\text{O}_2/\text{H}_2\text{O}} = +1.223 \text{ V}$$

Will the permanganate ion, MnO_4^- liberate O_2 from water in the presence of an acid?

- No because $E^\circ_{\text{cell}} = -2.733 \text{ V}$
 - Yes, because $E^\circ_{\text{cell}} = +0.287 \text{ V}$
 - No, because $E^\circ_{\text{cell}} = -0.287 \text{ V}$
 - Yes, because $E^\circ_{\text{cell}} = +2.733 \text{ V}$
2. Consider the change in oxidation state of Bromine corresponding to different emf values as shown in the diagram below : (2018)



Then the species undergoing disproportionation is

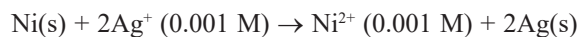
- BrO_3^-
 - BrO_4^-
 - HBrO
 - Br_2
3. A button cell used in watches functions as following
 $\text{Zn(s)} + \text{Ag}_2\text{O(s)} + \text{H}_2\text{O(l)} \rightleftharpoons 2\text{Ag(s)} + \text{Zn}^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq})$. If half cell potentials are
 $\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Zn(s)} ; E^\circ = -0.76 \text{ V}$
 $\text{Ag}_2\text{O(s)} + \text{H}_2\text{O(l)} + 2\text{e}^- \rightarrow 2\text{Ag(s)} + 2\text{OH}^-(\text{aq}) ; E^\circ = 0.34 \text{ V}$.
 The cell potential will be: (2013)
- 1.10 V
 - 0.42 V
 - 0.84 V
 - 1.34 V

Nernst Equation

4. At 298 K, the standard electrode potentials of Cu^{2+}/Cu , Zn^{2+}/Zn , Fe^{2+}/Fe and Ag^+/Ag are 0.34 V, -0.76 V, -0.44 V and 0.80 V, respectively.

On the basis of standard electrode potential, predict which of the following reaction can not occur? (2022)

- $2\text{CuSO}_4(\text{aq}) + 2\text{Ag(s)} \rightarrow 2\text{Cu(s)} + \text{Ag}_2\text{SO}_4(\text{aq})$
 - $\text{CuSO}_4(\text{aq}) + \text{Zn(s)} \rightarrow \text{ZnSO}_4(\text{aq}) + \text{Cu(s)}$
 - $\text{CuSO}_4(\text{aq}) + \text{Fe(s)} \rightarrow \text{FeSO}_4(\text{aq}) + \text{Cu(s)}$
 - $\text{FeSO}_4(\text{aq}) + \text{Zn(s)} \rightarrow \text{ZnSO}_4(\text{aq}) + \text{Fe(s)}$
5. Find the emf of the cell in which the following reaction takes place at 298 K



(Given that $E^\circ_{\text{cell}} = 10.5 \text{ V}$, $\frac{2.303 RT}{F} = 0.059$ at 298 K) (2022)

- 1.05 V
 - 1.0385 V
 - 1.385 V
 - 0.9615 V
6. Identify the reaction from following having top position in EMF series (Std. red. potential) according to their electrode potential at 298 K. (2020-Covid)
- $\text{Fe}^{2+} + 2\text{e}^- \rightarrow \text{Fe(s)}$
 - $\text{Au}^{3+} + 3\text{e}^- \rightarrow \text{Au(s)}$
 - $\text{K}^+ + 1\text{e}^- \rightarrow \text{K(s)}$
 - $\text{Mg}^{2+} + 2\text{e}^- \rightarrow \text{Mg(s)}$
7. For a cell involving one electron $E^\circ_{\text{cell}} = 0.59 \text{ V}$ at 298 K, the equilibrium constant for the cell reaction is:
 [Given that $\frac{2.303 RT}{F} = 0.059 \text{ V}$ at $T = 298 \text{ K}$] (2019)
- 1.0×10^2
 - 1.0×10^5
 - 1.0×10^{10}
 - 1.0×10^{30}
8. For the cell reaction
 $2\text{Fe}^{3+}(\text{aq}) + 2\text{I}^-(\text{aq}) \rightarrow 2\text{Fe}^{2+}(\text{aq}) + \text{I}_2(\text{aq})$
 $E^\circ_{\text{cell}} = 0.24 \text{ V}$ at 298 K. The standard Gibbs energy ($\Delta_r G^\circ$) of the cell reaction is:
 [Given that Faraday constant $F = 96500 \text{ C mol}^{-1}$] (2019)
- 46.32 kJ mol⁻¹
 - 23.16 kJ mol⁻¹
 - 46.32 kJ mol⁻¹
 - 23.16 kJ mol⁻¹

9. If the E°_{cell} for a given reaction has a negative value, which of the following gives the correct relationships for the values of ΔG° and K_{eq} ? (2016 - II)
- a. $\Delta G^\circ < 0$; $K_{\text{eq}} > 1$ b. $\Delta G^\circ < 0$; $K_{\text{eq}} < 1$
 c. $\Delta G^\circ > 0$; $K_{\text{eq}} < 1$ d. $\Delta G^\circ > 0$; $K_{\text{eq}} > 1$
10. The pressure of H_2 required to make the potential of H_2 electrode zero in pure water at 298 K is: (2016 - I)
- a. 10^{-4} atm b. 10^{-14} atm
 c. 10^{-12} atm d. 10^{-10} atm
11. The pair of compounds that can exist together is: (2014)
- a. HgCl_2 , SnCl_2 b. FeCl_2 , SnCl_2
 c. FeCl_3 , KI d. FeCl_3 , SnCl_2
12. A hydrogen gas electrode is made by dipping platinum wire in a solution of HCl of $\text{pH} = 10$ and by passing hydrogen gas around the platinum wire at one atm pressure. The oxidation potential of electrode would be? (2013)
- a. 0.059 V b. 0.59 V
 c. 0.118 V d. 1.18 V

Conductance of Electrolytic Solutions

13. The molar conductance of NaCl , HCl and CH_3COONa at infinite dilution are 126.45, 426.16 and $91.0 \text{ S cm}^2 \text{ mol}^{-1}$ respectively. The molar conductance of CH_3COOH at infinite dilution is. Choose the right option for your answer. (2021)
- a. $390.71 \text{ S cm}^2 \text{ mol}^{-1}$ b. $698.28 \text{ S cm}^2 \text{ mol}^{-1}$
 c. $540.48 \text{ S cm}^2 \text{ mol}^{-1}$ d. $201.28 \text{ S cm}^2 \text{ mol}^{-1}$
14. The molar conductivity of 0.007 M acetic acid is $20 \text{ S cm}^2 \text{ mol}^{-1}$. What is the dissociation constant of acetic acid? Choose the correct option. (2021)
- $$\left[\begin{array}{l} \Lambda^\circ_{\text{H}^+} = 350 \text{ S cm}^2 \text{ mol}^{-1} \\ \Lambda^\circ_{\text{CH}_3\text{COO}^-} = 50 \text{ S cm}^2 \text{ mol}^{-1} \end{array} \right]$$
- a. $2.50 \times 10^{-4} \text{ mol L}^{-1}$ b. $1.75 \times 10^{-5} \text{ mol L}^{-1}$
 c. $2.50 \times 10^{-5} \text{ mol L}^{-1}$ d. $1.75 \times 10^{-4} \text{ mol L}^{-1}$
15. The molar conductivity of a 0.5 mol dm^{-3} solution of AgNO_3 with electrolytic conductivity of $5.76 \times 10^{-3} \text{ S cm}^{-1}$ at 298 K is: (2016 - II)
- a. $0.086 \text{ S cm}^2 \text{ mol}^{-1}$ b. $28.8 \text{ S cm}^2 \text{ mol}^{-1}$
 c. $2.88 \text{ S cm}^2 \text{ mol}^{-1}$ d. $11.52 \text{ S cm}^2 \text{ mol}^{-1}$
16. At 25°C , molar conductance of 0.1 molar aqueous solution of ammonium hydroxide is $9.54 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$ and at infinite dilution its molar conductance is $238 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$. The degree of ionisation of ammonium hydroxide at the same concentration and temperature is: (2013)
- a. 2.080 % b. 20.800 %
 c. 4.008 % d. 40.800 %

Electrolytic Cells and Electrolysis

17. On electrolysis of dil sulphuric acid using Platinum (Pt) electrode, the product obtained at anode will be: (2020)
- a. Oxygen gas b. H_2S gas
 c. SO_2 gas d. Hydrogen gas
18. The number of Faradays (F) required to produce 20 g of calcium from molten CaCl_2 (Atomic mass of $\text{Ca} = 40 \text{ g mol}^{-1}$) is: (2020)
- a. 2 b. 3
 c. 4 d. 1
19. During the electrolysis of molten sodium chloride, the time required to produce 0.10 mol of chlorine gas using a current of 3 amperes is: (2016 - II)
- a. 220 minutes b. 330 minutes
 c. 55 minutes d. 110 minutes
20. The number of electrons delivered at the cathode during electrolysis by a current of 1 ampere in 60 seconds is: (charge on electron = $1.60 \times 10^{-19} \text{ C}$) (2016 - II)
- a. 3.75×10^{20} b. 7.48×10^{23}
 c. 6×10^{23} d. 6×10^{20}
21. When 0.1 mol MnO_4^{2-} is oxidised, the quantity of electricity required to completely oxidise MnO_4^{2-} to MnO_4^- is: (2014)
- a. 96500 C b. $2 \times 96500 \text{ C}$
 c. 9650 C d. 96.50 C
22. The weight of silver (atomic weight = 108) displaced by a quantity of electricity which displaces 5600 mL of O_2 at STP will be: (2014)
- a. 10.8 g b. 54.0 g
 c. 108.0 g d. 5.4 g

Batteries, Fuel Cells and Corrosion

23. In a typical fuel cell, the reactants (R) and product (P) are (2020-Covid)
- a. $\text{R} = \text{H}_{2(\text{g})}$, $\text{O}_{2(\text{g})}$; $\text{P} = \text{H}_2\text{O}_{(\text{l})}$
 b. $\text{R} = \text{H}_{2(\text{g})}$, $\text{O}_{2(\text{g})}$, $\text{Cl}_{2(\text{g})}$; $\text{P} = \text{HClO}_{4(\text{aq})}$
 c. $\text{R} = \text{H}_{2(\text{g})}$, $\text{N}_{2(\text{g})}$; $\text{P} = \text{NH}_{3(\text{aq})}$
 d. $\text{R} = \text{H}_{2(\text{g})}$, $\text{O}_{2(\text{g})}$; $\text{P} = \text{H}_2\text{O}_{2(\text{l})}$
24. Zinc can be coated on iron to produce galvanized iron but the reverse is not possible. It is because: (2016 - II)
- a. Zinc has lower negative electrode potential than iron
 b. Zinc has higher negative electrode potential than iron
 c. Zinc is lighter than iron
 d. Zinc has lower melting point than iron
25. A device that converts energy of combustion of fuels like hydrogen and methane, directly into electrical energy is known as: (2015)
- a. Electrolytic cell b. Dynamo
 c. Ni-Cd cell d. Fuel cell

Answer Key

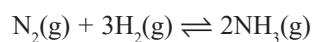
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
b	c	a	a	None	b	c	a	c	b	b	b	a	b	d	c	a
18	19	20	21	22	23	24	25									
d	d	a	c	c	a	b	d									

CHAPTER 13

Chemical Kinetics

Rate of Chemical Reactions

1. For the chemical reaction



The correct option is:

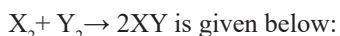
(2019)

- a. $-\frac{1}{3} \frac{d[\text{H}_2]}{dt} = -\frac{1}{2} \frac{d[\text{NH}_3]}{dt}$ b. $-\frac{d[\text{N}_2]}{dt} = 2 \frac{d[\text{NH}_3]}{dt}$
c. $-\frac{d[\text{N}_2]}{dt} = \frac{1}{2} \frac{d[\text{NH}_3]}{dt}$ d. $3 \frac{d[\text{H}_2]}{dt} = 2 \frac{d[\text{NH}_3]}{dt}$

Factors Influencing Rate of a Reaction & Order of Reaction

2. Mechanism of a hypothetical reaction

(2017-Delhi)



- (i) $\text{X}_2 \rightarrow \text{X} + \text{X}$ (fast)
(ii) $\text{X} + \text{Y}_2 \rightleftharpoons \text{XY} + \text{Y}$ (slow)
(iii) $\text{X} + \text{Y} \rightarrow \text{XY}$ (fast)

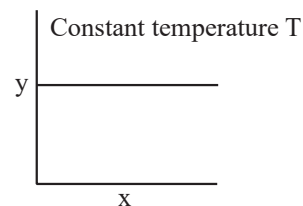
The overall order of the reaction will be

- a. 1.5 b. 1
c. 2 d. 0
3. The decomposition of phosphine (PH_3) on tungsten at low pressure is a first-order reaction. It is because the: (2016-II)
- a. Rate is proportional to the surface coverage
b. Rate is inversely proportional to the surface coverage
c. Rate is independent of the surface coverage
d. Rate of decomposition is very low

Integrated Rate Equation & Half Life of Reactions

4. For a first order reaction $\text{A} \rightarrow \text{Products}$, initial concentration of A is 0.1 M, which becomes 0.001 M after 5 minutes. Rate constant for the reaction in min^{-1} is (2022)
- a. 0.2303 b. 1.3818
c. 0.9212 d. 0.4606

5. The given graph is a representation of kinetics of a reaction.



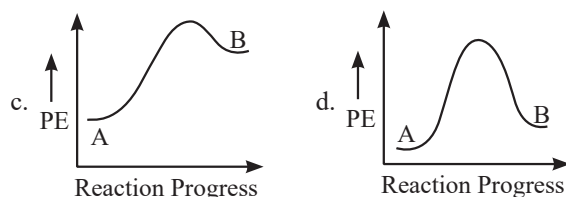
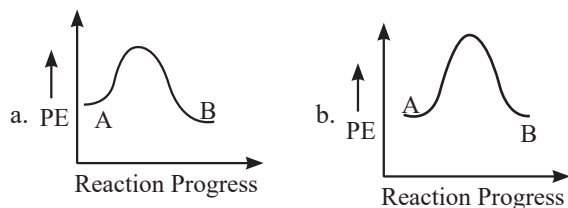
The y and x axes for zero and first order reactions, respectively are (2022)

- a. zero order ($y = \text{rate}$ and $x = \text{concentration}$), first order ($y = \text{rate}$ and $x = t_{1/2}$)
b. zero order ($y = \text{concentration}$ and $x = \text{time}$), first order ($y = t_{1/2}$ and $x = c$ concentration)
c. zero order ($y = \text{concentration}$ and $x = \text{time}$), first order ($y = \text{rate constant}$ and $x = \text{concentration}$)
d. zero order ($y = \text{rate}$ and $x = \text{concentration}$), first order ($y = t_{1/2}$ and $x = \text{concentration}$)
6. The rate constant for a first order reaction is $4.606 \times 10^{-3} \text{ s}^{-1}$. The time required to reduce 2.0 g of the reactant to 0.2 g is: (2020)
- a. 200 s b. 500 s
c. 1000 s d. 100 s
7. The half-life for a zero order reaction having 0.02 M initial concentration of reactant is 100 s. The rate constant (in $\text{mol L}^{-1} \text{ s}^{-1}$) for the reaction is (2020-Covid)
- a. 2.0×10^{-4} b. 2.0×10^{-3}
c. 1.0×10^{-2} d. 1.0×10^{-4}
8. If the rate constant for a first order reaction is k, the time (t) required for the completion of 99% of the reaction is given by: (2019)
- a. $t = 0.693/k$
b. $t = 6.909/k$
c. $t = 4.606/k$
d. $t = 2.303/k$
9. When initial concentration of the reactant is doubled, the half-life period of a zero order reaction (2018)
- a. Is halved
b. Is doubled
c. Is tripled
d. Remains unchanged

10. The correct difference between first and second order reactions is that: (2018)
- The rate of a first-order reaction does not depend on reactant concentrations; the rate of a second-order reaction does depend on reactant concentrations
 - The half-life of a first-order reaction does not depend on $[A]_0$; the half-life of a second-order reaction does depend on $[A]_0$
 - The rate of a first-order reaction does depend on reactant concentrations; the rate of a second-order reaction does not depend on reactant concentrations
 - A first-order reaction can catalyzed; a second-order reaction cannot be catalyzed
11. A first order reaction has a specific reaction rate of 10^{-2} s^{-1} . How much time will it take for 20 g of the reactant to reduce to 5 g? (2017-Delhi)
- 693.0 second
 - 238.6 second
 - 138.6 second
 - 346.5 second
12. The rate of a first-order reaction is $0.04 \text{ mol L}^{-1} \text{ s}^{-1}$ at 10 seconds and $0.03 \text{ mol L}^{-1} \text{ s}^{-1}$ at 20 seconds after initiation of the reaction. The half-life period of the reaction is: (2016 - I)
- 54.1 s
 - 24.1 s
 - 34.1 s
 - 44.1 s
13. The rate constant of the reaction $A \rightarrow B$ is $0.6 \times 10^{-3} \text{ mole per second}$. If the concentration of A is 5 M, then concentration of B after 20 minutes is: (2015 Re)
- 0.72 M
 - 1.08 M
 - 3.60 M
 - 0.36 M
14. When initial concentration of a reactant is doubled in a reaction, its half-life period is not affected. The order of the reaction is: (2015)
- First
 - Second
 - More than zero but less than first
 - Zero

Temperature Dependence of the Rate of a Reaction & Effect of Catalysts

15. For a reaction $A \rightarrow B$, enthalpy of reaction is -4.2 kJ mol^{-1} and enthalpy of activation is 9.6 kJ mol^{-1} . The correct potential energy profile for the reaction is shown in option. (2021)



16. The slope of Arrhenius Plot ($\ln K$ v/s $1/T$) of first order reaction is $-5 \times 10^3 \text{ K}$. The value of E_a of the reaction is. Choose the correct option for your answer. (2021)
[Given $R = 8.314 \text{ JK}^{-1} \text{ mol}^{-1}$]
- 83.0 kJ mol^{-1}
 - 166 kJ mol^{-1}
 - -83 kJ mol^{-1}
 - 41.5 kJ mol^{-1}
17. The addition of a catalyst during a chemical reaction alters which of the following quantities? (2016 - I)
- Activation energy
 - Entropy
 - Internal energy
 - Enthalpy
18. The activation energy of a reaction can be determined from the slope of which of the following graphs? (2015)
- $\frac{\ln k}{T}$ vs. T
 - $\ln k$ vs. $\frac{1}{T}$
 - $\frac{T}{\ln k}$ vs. $\frac{1}{T}$
 - $\ln k$ vs. T
19. A reaction having equal energies of activation for forward and reverse reactions has: (2013)
- $\Delta S = 0$
 - $\Delta G = 0$
 - $\Delta H = 0$
 - $\Delta H = \Delta G = \Delta S = 0$
20. What is the activation energy for a reaction if its rate doubles when the temperature is raised from 20°C to 35°C ? ($R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$) (2013)
- 342 kJ mol^{-1}
 - 269 kJ mol^{-1}
 - 34.7 kJ mol^{-1}
 - 15.1 kJ mol^{-1}

Collision Theory of Chemical Reactions

21. In collision theory of chemical reaction, Z_{AB} represents (2020-Covid)
- The collision frequency of reactants, A and B
 - Steric factor
 - The fraction of molecules with energies equal to E_a
 - The fraction of molecules with energies greater than E_a
22. An increase in the concentration of the reactants of a reaction leads to change in : (2020)
- Heat of reaction
 - Threshold energy
 - Collision frequency
 - Activation energy

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
c	a	a	c	d	b	d	c	b	b	c	b	a	a	a	d	a
18	19	20	21	22												
b	c	c	a	c												

CHAPTER 14

Atomic Structure

Subatomic Particles, Atomic Models

1. The number of protons, neutrons and electrons in ${}^{175}_{71}\text{Lu}$, respectively, are : (2020)
- 104, 71 and 71
 - 71, 71 and 104
 - 175, 104 and 71
 - 71, 104 and 71

Developments Leading to Bohr's Model of Atom

2. A particular station of All India Radio, New Delhi, broadcasts on a frequency of 1,368 kHz (kilohertz). The wavelength of the electromagnetic radiation emitted by the transmitter is: [speed of light, $c = 3.0 \times 10^8 \text{ ms}^{-1}$] (2021)
- 219.2 m
 - 2192 m
 - 21.92 cm
 - 219.3 m
3. Which of the following series of transitions in the spectrum of hydrogen atom fall in visible region? (2019)
- Lyman series
 - Balmer series
 - Paschen series
 - Brackett series
4. Calculate the energy in joule corresponding to light of wavelength 45 nm (Planck's constant $h = 6.63 \times 10^{-34} \text{ Js}$; speed of light $c = 3 \times 10^8 \text{ ms}^{-1}$) (2014)
- 6.67×10^{11}
 - 4.42×10^{-15}
 - 4.42×10^{-18}
 - 6.67×10^{15}
5. The value of Planck's constant is $6.63 \times 10^{-34} \text{ Js}$. The speed of light is $3 \times 10^8 \text{ ms}^{-1}$. Which value is closest to the wavelength in nanometer of a quantum of light with frequency of $6 \times 10^{15} \text{ s}^{-1}$? (2013)
- 10 nm
 - 25 nm
 - 50 nm
 - 75 nm

Bohr's Model For Hydrogen Atom

6. If radius of second Bohr orbit of the He^+ ion is 105.8 pm, what is the radius of third Bohr orbit of Li^{2+} ion? (2022)
- 158.7 Å
 - 158.7 pm
 - 15.87 pm
 - 1.587 pm

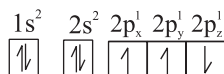
7. Based on equation, $E = -2.178 \times 10^{-18} \text{ J} \left(\frac{Z^2}{n^2} \right)$ certain conclusions are written. Which of them is not correct? (2013)

- The negative sign in equation simply means that the energy of electron bound to the nucleus is lower than it would be if the electrons were at the infinite distance from the nucleus.
- Larger the value of n , the larger is the orbit radius.
- Equation can be used to calculate the change in energy when the electron changes orbit.
- For $n = 1$, the electron has a more negative energy than it does for $n = 6$ which means that the electron is more loosely bound in the smallest allowed orbit.

Quantum Mechanical Model of Atom

8. Identify the incorrect statements from the following. (2022)
- The shapes of d_{xy} , d_{yz} , and d_{zx} orbitals are similar to each other; and $d_{x^2-y^2}$ and d_{z^2} are similar to each other.
 - All the five 5d orbitals are different in size when compared to the respective 4d orbitals.
 - All the five 4d orbitals have shapes similar to the respective 3d orbitals.
 - In an atom, all the five 3d orbitals are equal in energy in free state.
9. The number of angular nodes and radial nodes in 3s orbital are (2020-Covid)
- 1 and 0, respectively
 - 3 and 0, respectively
 - 0 and 1, respectively
 - 0 and 2, respectively
10. 4d, 5p, 5f and 6p orbitals are arranged in the order of decreasing energy. The correct option is (2019)
- $5f > 6p > 5p > 4d$
 - $6p > 5f > 5p > 4d$
 - $6p > 5f > 4d > 5p$
 - $5f > 6p > 4d > 5p$

11. Which one is a wrong statement? (2018)
- Total orbital angular momentum of electron in 's' orbital is equal to zero
 - An orbital is designated by three quantum numbers while an electron in an atom is designated by four quantum numbers
 - The value of m for d_z^2 is zero
 - The electronic configuration of N atom is



12. Which one is the wrong statement? (2017-Delhi)
- The energy of 2s orbital is less than the energy of 2p orbital in case of hydrogen like atoms
 - de-Broglie's wavelength is given by $\lambda = \frac{h}{mv}$, where m = mass of the particle, v = group velocity of the particle
 - The uncertainty principle is $\Delta E \times \Delta t \geq \frac{h}{4\pi}$
 - Half-filled and fully filled orbitals have greater stability due to greater exchange energy, greater symmetry and more balanced arrangement
13. Which of the following pairs of d-orbitals will have electron density along the axes? (2016 - II)
- $d_z, d_{x^2-y^2}$
 - $d_{xy}, d_{x^2-y^2}$
 - d_z, d_{xz}
 - d_{xz}, d_{yz}
14. How many electrons can fit in the orbital for which: $n = 3$ and $l = 1$? (2016 - II)

- 10
 - 2
 - 14
 - 6
15. Two electrons occupying the same orbital are distinguished by: (2016 - I)
- Spin quantum number
 - Principal quantum number
 - Magnetic quantum number
 - Azimuthal quantum number
16. Which is the correct order of increasing energy of the listed orbitals in the atom of titanium? (Atomic number $Z = 22$): (2015 Re)
- 3s 3p 4s 3d
 - 3s 4s 3p 3d
 - 4s 3s 3p 3d
 - 3s 3p 3d 4s
17. The angular momentum of electron in 'd' orbital is equal to: (2015)
- $2\sqrt{3}\hbar$
 - $0\hbar$
 - $\sqrt{6}\hbar$
 - $\sqrt{2}\hbar$
18. What is the maximum number of orbitals that can be identified with the following quantum numbers: $n = 3, l = 1, m_l = 0$? (2014)
- 2
 - 3
 - 4
 - 1
19. What is the maximum number of electrons that can be associated with the following set of quantum numbers: $n = 3, \ell = 1$ and $m = -1$. (2013)
- 10
 - 6
 - 4
 - 2

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
d	d	b	c	c	b	d	a	d	a	d	a	a	c	a	a	c
18	19															
d	d															

CHAPTER 15

Hydrocarbons

Alkanes

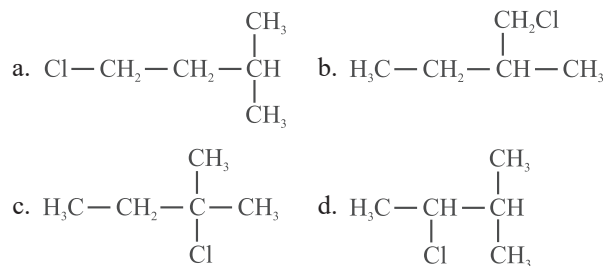
- Dihedral angle of least stable conformer of ethane is: (2021)
 - 180°
 - 60°
 - 0°
 - 120°
- Which of the following alkane cannot be made in good yield by Wurtz reaction? (2020)
 - 2, 3-Dimethylbutane
 - n*-Heptane
 - n*-Butane
 - n*-Hexane
- Which of the following is a free radical substitution reaction? (2020-Covid)
 - Acetylene with HBr
 - Methane with $\text{Br}_2/h\nu$
 - Propene with $\text{HBr}/(\text{C}_6\text{H}_5\text{COO})_2$
 - Benzene with $\text{Br}_2/\text{AlCl}_3$
- Hydrocarbon (A) reacts with bromine by substitution to form an alkyl bromide which by Wurtz reaction is converted to gaseous hydrocarbon containing less than four carbon atoms. (A) is (2018)
 - $\text{CH} \equiv \text{CH}$
 - $\text{CH}_2 = \text{CH}_2$
 - CH_4
 - $\text{CH}_3 - \text{CH}_3$
- With respect to the conformers of ethane, which of the following statements is true? (2017-Delhi)
 - Both bond angles and bond length remains same
 - Bond angle remains same but bond length changes
 - Bond angle changes but bond length remains same
 - Both bond angle and bond length change
- The correct statement regarding the comparison of staggered and eclipsed conformations of ethane, is: (2016 - I)
 - The staggered conformation of ethane is more stable than eclipsed conformation, because staggered conformation has no torsional strain.
 - The staggered conformation of ethane is less stable than eclipsed conformation, because staggered conformation has torsional strain.
 - The eclipsed conformation of ethane is more stable than staggered conformation, because eclipsed conformation has no torsional strain.
 - The eclipsed conformation of ethane is more stable than staggered conformation even though the eclipsed conformation has torsional strain.
- The number of structural isomers possible from the molecular formula $\text{C}_3\text{H}_9\text{N}$ is: (2015 Re)
 - 3
 - 4
 - 5
 - 2

Alkenes

- Compound X on reaction with O_3 followed by $\text{Zn}/\text{H}_2\text{O}$ gives formaldehyde and 2-methyl propanal as products. The compound X is: (2022)
 - Pent-2-ene
 - 3-Methylbut-1-ene
 - 2-Methylbut-1-ene
 - 2-Methylbut-2-ene
- The major product of the following chemical reaction is: (2021)

$$\begin{array}{c} \text{CH}_3 \\ \diagdown \\ \text{CH} - \text{CH} = \text{CH}_2 \\ \diagup \\ \text{CH}_3 \end{array} + \text{HBr} \xrightarrow{(\text{C}_6\text{H}_5\text{CO})_2\text{O}_2} ?$$
 - $\begin{array}{c} \text{CH}_3 \\ \diagdown \\ \text{CH} - \text{CH}_2 - \text{CH}_2 - \text{O} - \text{COC}_6\text{H}_5 \\ \diagup \\ \text{CH}_3 \end{array}$
 - $\begin{array}{c} \text{CH}_3 \\ \diagdown \\ \text{CH} - \text{CH} - \text{CH}_3 \\ \diagup \quad | \\ \text{CH}_3 \quad \text{Br} \end{array}$
 - $\begin{array}{c} \text{CH}_3 \\ \diagdown \\ \text{CBr} - \text{CH}_2 - \text{CH}_3 \\ \diagup \\ \text{CH}_3 \end{array}$
 - $\begin{array}{c} \text{CH}_3 \\ \diagdown \\ \text{CH} - \text{CH}_2 - \text{CH}_2 - \text{Br} \\ \diagup \\ \text{CH}_3 \end{array}$
- An alkene on ozonolysis gives methanal as one of the product. Its structure is : (2020)
 - $\begin{array}{c} \text{CH}_2 - \text{CH}_2 - \text{CH}_3 \\ | \\ \text{Cyclohexene} \end{array}$
 - $\begin{array}{c} \text{CH}_2 - \text{CH} = \text{CH}_2 \\ | \\ \text{Cyclohexene} \end{array}$
 - $\begin{array}{c} \text{CH}_2\text{CH}_2\text{CH}_3 \\ | \\ \text{Cyclohexene} \end{array}$
 - $\begin{array}{c} \text{CH} = \text{CH} - \text{CH}_3 \\ | \\ \text{Cyclohexene} \end{array}$

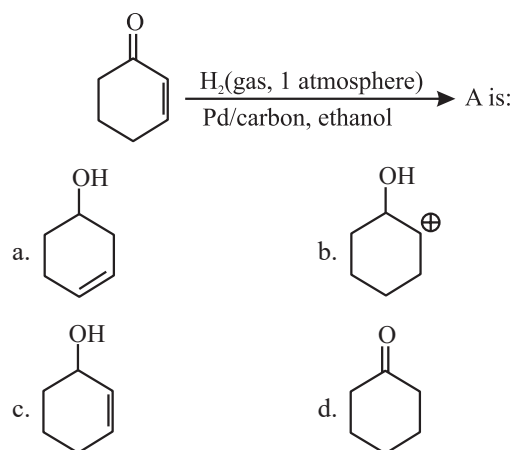
11. An alkene "A" on reaction with O_3 and $Zn - H_2O$ gives propanone and ethanal in equimolar ratio. Addition of HCl to alkene "A" gives "B" as the major product. The structure of product "B" is: (2019)



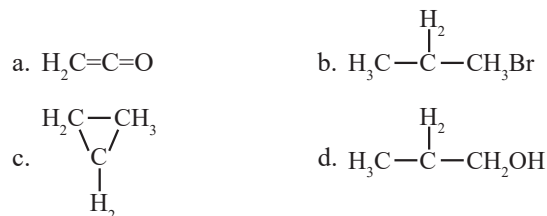
12. The compound that will react most readily with gaseous bromine has the formula: (2016 - II)



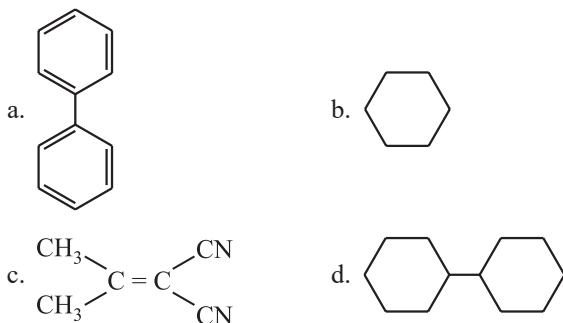
13. The correct structure of the product A formed in the reaction: (2016 - II)



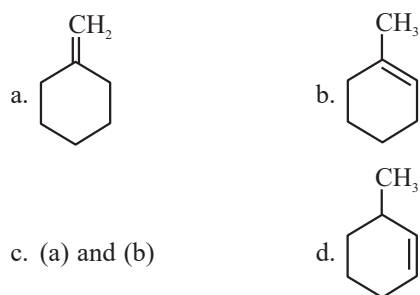
14. Which of the following compounds shall not produce propene by reaction with HBr followed by elimination or direct only elimination reaction? (2016 - II)



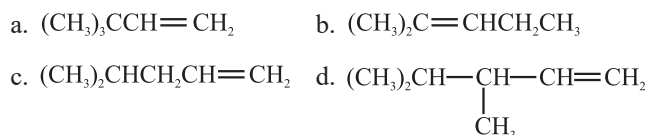
15. In which of the following molecules, all atoms are coplanar? (2016 - II)



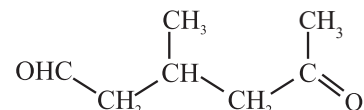
16. In the reaction with HCl, an alkene reacts in accordance with the Markovnikov's rule, to give a product 1-chloro-1-methylcyclohexane. The possible alkene is: (2015 Re)



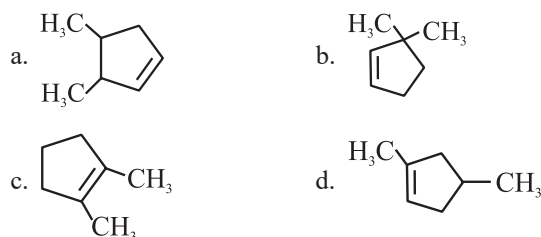
17. 2, 3-Dimethyl-2-butene can be prepared by heating which of the following compounds with a strong acid? (2015 Re)



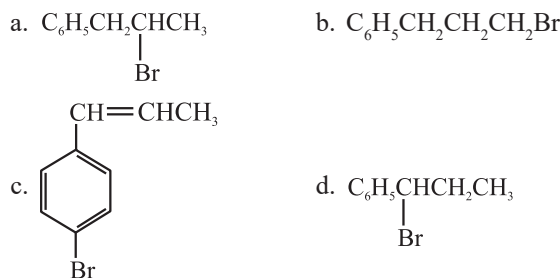
18. A single compound of the structure



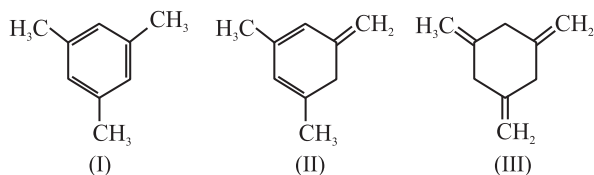
is obtainable from ozonolysis of which of the following cyclic compounds? (2015)



19. The reaction of $C_6H_5CH=CHCH_3$ with HBr produces: (2015)



20. Given:

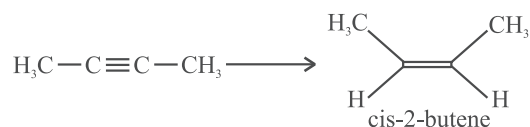


The enthalpy of hydrogenation of these compounds will be in the order as: (2015)

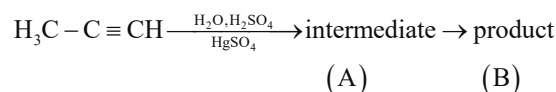
- a. III > II > I b. II > III > I
c. II > I > III d. I > II > III

Alkynes

21. The number of sigma (s) and pi (p) bonds in pent-2-en-4-yne is
(2019)
- 10 s bonds and 3p bonds
 - 8 s bonds and 5p bonds
 - 11 s bonds and 2p bonds
 - 13 s bonds and no p bonds
22. The most suitable reagent for the following conversion, is:
(2019)



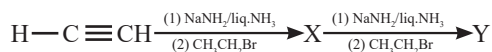
- Na/liquid NH_3
 - H_2 , Pd/C, quinoline
 - Zn/HCl
 - $\text{Hg}^{2+}/\text{H}^+$, H_2O
23. Predict the correct intermediate and product in the following reaction
(2017-Delhi)



- A: $\text{H}_3\text{C}-\text{C}(\text{SO}_4)=\text{CH}_2$ B: $\text{H}_3\text{C}-\text{C}(=\text{O})-\text{CH}_3$
- A: $\text{H}_3\text{C}-\text{C}(\text{OH})=\text{CH}_2$ B: $\text{H}_3\text{C}-\text{C}(=\text{O})-\text{CH}_3$
- A: $\text{H}_3\text{C}-\text{C}(\text{OH})=\text{CH}_2$ B: $\text{H}_3\text{C}-\text{C}(\text{SO}_4)=\text{CH}_2$
- A: $\text{H}_3\text{C}-\text{C}(=\text{O})-\text{CH}_3$ B: $\text{H}_3\text{C}-\text{C}\equiv\text{CH}$

24. Which one is the correct order of acidity? (2017-Delhi)
- $\text{CH}_3-\text{CH}_3 > \text{CH}_2=\text{CH}_2 > \text{CH}_3-\text{C}\equiv\text{CH} > \text{CH}\equiv\text{CH}$
 - $\text{CH}_2=\text{CH}_2 > \text{CH}_3-\text{CH}=\text{CH}_2 > \text{CH}_3-\text{C}\equiv\text{CH} > \text{CH}\equiv\text{CH}$
 - $\text{CH}\equiv\text{CH} > \text{CH}_3-\text{C}\equiv\text{CH} > \text{CH}_2=\text{CH}_2 > \text{CH}_3-\text{CH}_3$
 - $\text{CH}\equiv\text{CH} > \text{CH}_2=\text{CH}_2 > \text{CH}_3-\text{C}\equiv\text{CH} > \text{CH}_3-\text{CH}_3$

25. In the reaction: (2016 - I)



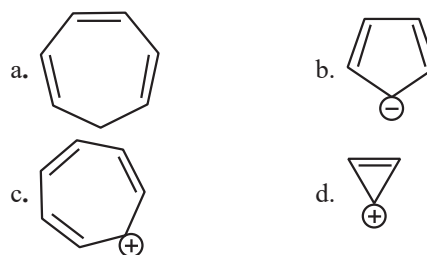
- X = 1-Butyne ; Y = 2-Hexyne
- X = 1-Butyne ; Y = 3-Hexyne
- X = 2-Butyne ; Y = 3-Hexyne
- X = 2-Butyne ; Y = 2-Hexyne

26. The pair of electron in the given carbanion, $\text{CH}_3\text{C}\equiv\text{C}^-$, is present in which of the following orbitals? (2016 - I)

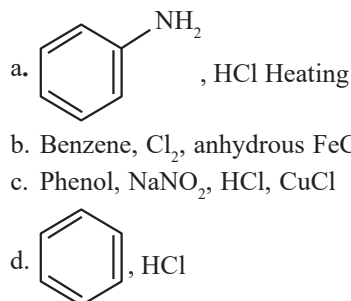
- sp
- d^2sp
- sp^3
- sp^2

Aromatic Hydrocarbons

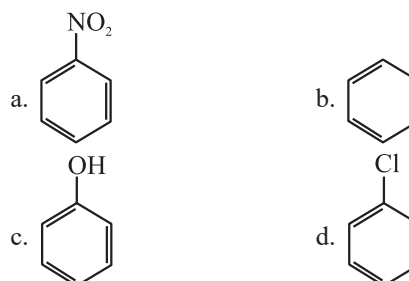
27. Which compound amongst the following is not an aromatic compound? (2022)



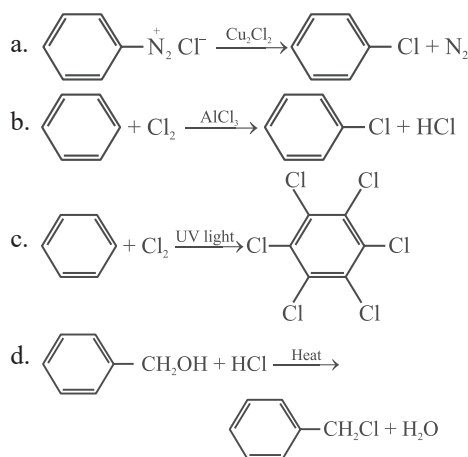
28. Which of the following is suitable to synthesize chlorobenzene? (2022)



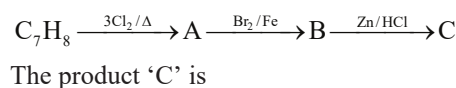
29. Which of the following compound is most reactive in electrophilic aromatic substitution? (2020-Covid)



30. Among the following, the reaction that proceeds through an electrophilic substitution, is: (2019)



31. The compound C_7H_8 undergoes the following reactions: (2018)

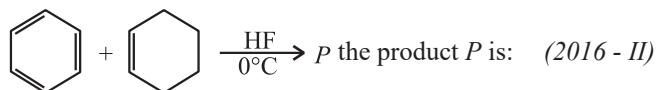


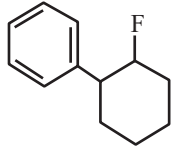
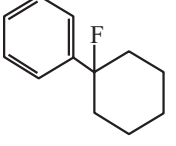
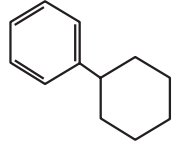
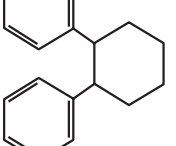
- a. m-bromotoluene
- b. o-bromotoluene
- c. p-bromotoluene
- d. 3-bromo-2,4,6-trichlorotoluene

32. Which of the following can be used as the halide component for Friedel-Crafts reaction ? (2016 - II)

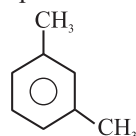
- a. Chloroethene
- b. Isopropyl chloride
- c. Chlorobenzene
- d. Bromobenzene

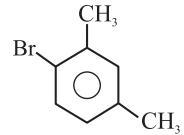
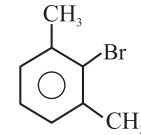
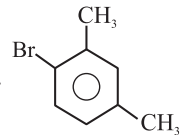
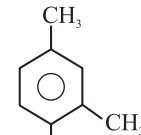
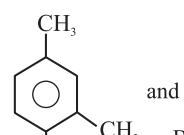
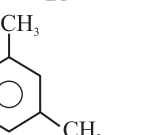
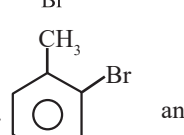
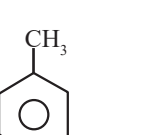
33. In the given reaction,



- a. 
- b. 
- c. 
- d. 

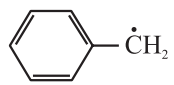
34. What products are formed when the following compound is treated with Br_2 in the presence of FeBr_3 ? (2014)



- a.  and 
- b.  and 
- c.  and 
- d.  and 

35. Which of the following compounds will not undergo Friedel-Craft reaction easily? (2013)

- a. Cumene
- b. Xylene
- c. Nitrobenzene
- d. Toluene

36. The radical,  is aromatic because it has: (2013)

- a. 6 p-orbitals and 6 unpaired electrons
- b. 7 p-orbitals and 6 unpaired electrons
- c. 7 p-orbitals and 7 unpaired electrons
- d. 6 p-orbitals and 7 unpaired electrons

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
c	b	b	c	a	a	b	b	d	b	c	c	d	a	a	c	a
18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
d	d	a	a	b	b	c	b	a	a	b	c	b	a	b	c	b
35	36															
c	a															

CHAPTER 16

Amines

Preparation of Amines

- Reaction of propanamide with ethanolic sodium hydroxide and bromine will give (2020-Covid)
 - Methylamine
 - Propylamine
 - Aniline
 - Ethylamine
- Method by which Aniline cannot be prepared is: (2015 Re)
 - Potassium salt of phthalimide treated with chlorobenzene followed by hydrolysis with aqueous NaOH solution
 - Hydrolysis of phenylisocyanide with acidic solution
 - Degradation of benzamide with bromine in alkaline solution
 - Reduction of nitrobenzene with H_2/Pd in ethanol

Physical & Chemical Properties of Amines

- Given below are two statements: (2022)

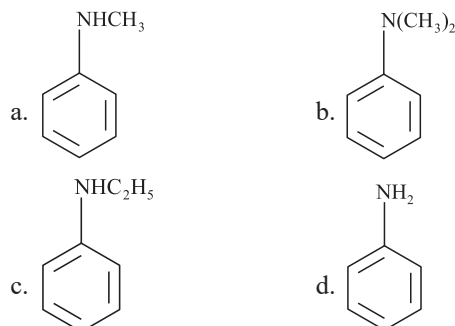
Statements I : Primary aliphatic amines react with HNO_2 to give unstable diazonium salts.

Statements II : Primary aromatic amines react with HNO_2 to form diazonium salts which are stable even above 300 K.

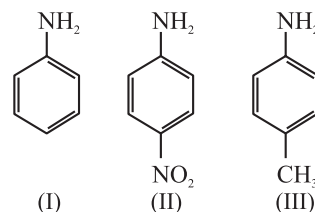
In the light of the above Statements, choose the most appropriate answer from the options given below:

 - Statements I is incorrect but Statements II is correct.
 - Both Statements I and Statements II are correct
 - Both Statements I and Statements II are incorrect.
 - Statements I is correct but Statement II is incorrect.
- Identify the compound that will react with Hinsberg's reagent to give a solid which dissolves in alkali. (2021)
 -
 -
 -
 -

- Which of the following amine will give the carbylamine test? (2020)



- The correct order of the basic strength of methyl substituted amines in aqueous solution is: (2019)
 - $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N$
 - $(CH_3)_3N > CH_3NH_2 > (CH_3)_2NH$
 - $(CH_3)_3N > (CH_3)_2NH > CH_3NH_2$
 - $CH_3NH_2 > (CH_3)_2NH > (CH_3)_3N$
- Nitration of aniline in strong acidic medium also gives m-nitroaniline because: (2018)
 - In spite of substituents nitro group always goes to only m-position.
 - In electrophilic substitution reactions, amino group is meta directive.
 - In acidic (strong) medium, aniline is present as anilinium ion.
 - In absence of substituents nitro group always goes to m-position.
- The correct increasing order of basic strength for the following compounds is: (2017-Delhi)

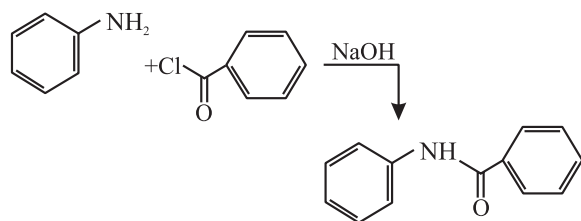


- $II < I < III$
- $II < III < I$
- $III < I < II$
- $III < II < I$

9. The correct statement regarding the basicity of arylamines is:
(2016 - I)

- Arylamines are generally more basic than alkylamines, because the nitrogen atom in arylamines is sp -hybridised.
- Arylamines are generally less basic than alkylamines because the nitrogen lone pair electrons are delocalized by interaction with the aromatic ring π electrons system.
- Arylamines are generally more basic than alkylamines because the nitrogen lone pair electrons are not delocalised by interaction with the aromatic ring π electron system.
- Arylamines are generally more basic than alkylamines because of aryl group

10. The following reaction



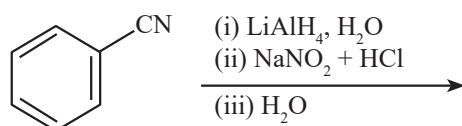
is known by the name:

[OS] (2015 Re)

- Schotten-Baumen reaction
- Friedel-Craft's reaction
- Perkin's reaction
- Acetylation reaction

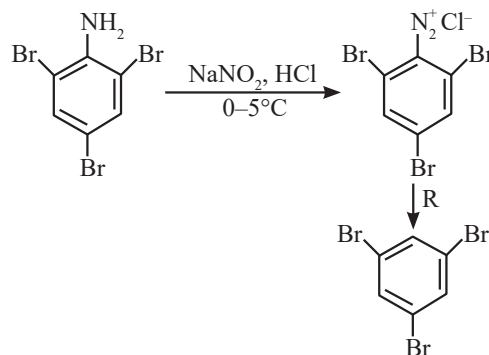
Diazonium Salt (Preparation, Physical & Chemical Properties)

11. The product formed from the following reaction sequence is
(2022)



-
-
-
-

12. The reagent 'R' in the given sequence of chemical reaction is:
(2021)

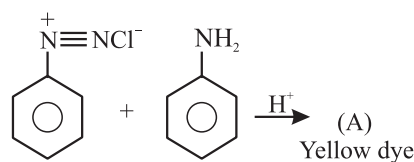


- CH_3CH_2OH
- HI
- $CuCN/KCN$
- H_2O

13. A given nitrogen-containing aromatic compound A reacts with Sn/HCl , followed by HNO_2 to give an unstable compound B. B, on treatment with phenol, forms a beautiful coloured compound C with the molecular formula $C_{12}H_{10}N_2O$. The structure of compound A is:
(2016 - II)

-
-
-
-

14. In the following reaction, the product (A) is:
(2014)

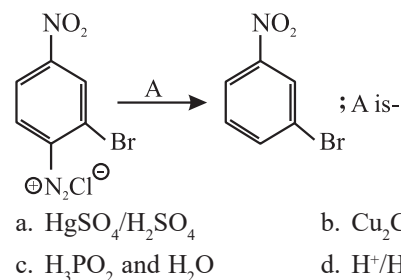


-
-
-
-

15. Which of the following will be most stable diazonium salt $RN_2^+X^-$?
(2014)

- $C_6H_5N_2^+X^-$
- $CH_3CH_2N_2^+X^-$
- $C_6H_5CH_2N_2^+H^-$
- $CH_3N_2^+X^-$

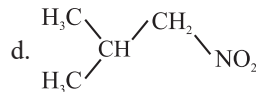
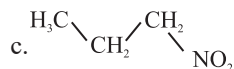
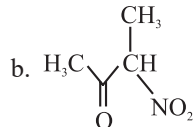
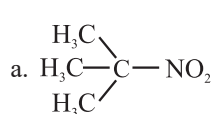
16. In the reaction:
(2013)



- $HgSO_4/H_2SO_4$
- Cu_2Cl_2
- H_3PO_2 and H_2O
- H^+/H_2O

Nitrobenzene (Preparation and Properties)

17. Which one of the following nitro-compounds does not react with nitrous acid? (2016 - II)



18. The electrolytic reduction of nitrobenzene in strongly acidic medium produces (2015)

- a. Azobenzene
- b. Aniline
- c. p-aminophenol
- d. Azoxybenzene

19. Nitrobenzene on reaction with conc. $\text{HNO}_3/\text{H}_2\text{SO}_4$ at $80^\circ - 100^\circ\text{C}$ forms which one of the following products? (2013)

- a. 1,2-Dinitrobenzene
- b. 1,3-Dinitrobenzene
- c. 1,4-Dinitrobenzene
- d. 1,2,4-Trinitrobenzene

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
d	a	d	b	d	a	c	a	b	a	a	a	d	c	a	c	a
18	19															
c	b															

CHAPTER 17

Alcohols, Phenols and Ethers

Alcohols

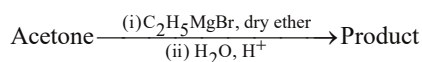
1. Given below are two statements: (2022)

Statement I: In Lucas test, primary, secondary and tertiary alcohols are distinguished on the basis of their reactivity with conc. $\text{HCl} + \text{ZnCl}_2$, known as Lucas Reagent.

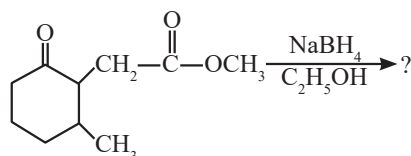
Statement II: Primary alcohols are most reactive and immediately produce turbidity at room temperature on reaction with Lucas Reagent.

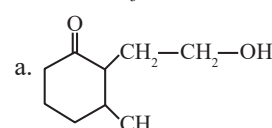
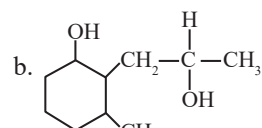
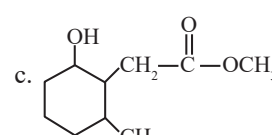
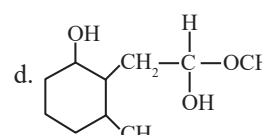
In the light of the above statements, choose the most appropriate answer from the options given below:

- Statement I is incorrect but Statement II is correct.
 - Both Statement I and Statement II are correct.
 - Both Statement I and Statement II are incorrect.
 - Statement I is correct but Statement II is incorrect.
2. What is the IUPAC name of the organic compound formed in the following chemical reaction? (2021)

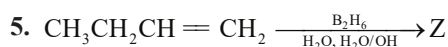


- Pentan-2-ol
 - Pentan-3-ol
 - 2-methyl butan-2-ol
 - 2-methyl propan-2-ol
3. The product formed in the following chemical reaction is: (2021)



- 
 - 
 - 
 - 
4. Reaction between acetone and methylmagnesium chloride followed by hydrolysis will give : (2020)

- Sec. butyl alcohol
- Tert. butyl alcohol
- Isobutyl alcohol
- Isopropyl alcohol

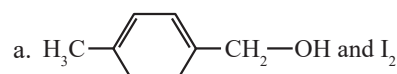
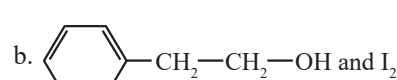
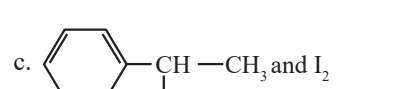
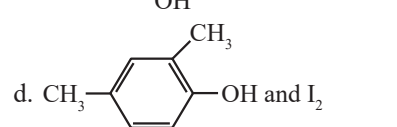


What is Z?

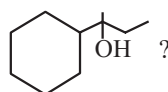
(2020-Covid)

- $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$
- $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$
- $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$
- $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$

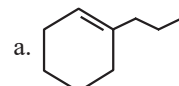
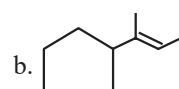
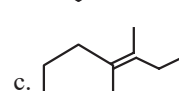
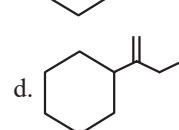
6. Compound A, $\text{C}_8\text{H}_{10}\text{O}$, is found to react with NaOI (produced by reacting Y with NaOH) and yields a yellow precipitate with characteristic smell. A and Y are respectively (2018)

- 
- 
- 
- 

7. Which of the following is not the product of dehydration of



(2015 Re)

- 
- 
- 
- 

Phenols

8. Given below are two statements: (2022)

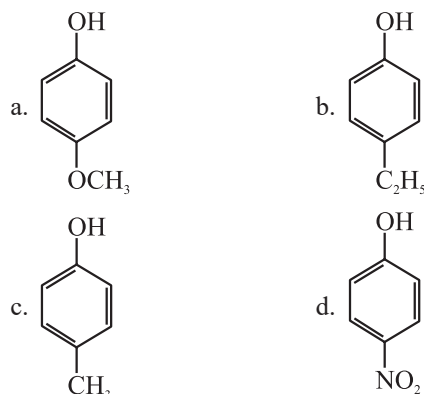
Statement I: The acidic strength of monosubstituted nitrophenol is higher than phenol because of electron withdrawing nitro group.

Statement II: o-nitrophenol, m-nitrophenol and p-nitrophenol will have same acidic strength as they have one nitro group attached to the phenolic ring.

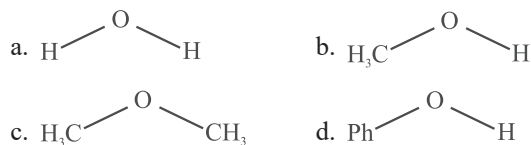
In the light of the above statements, choose the most appropriate answer form the options given below:

- Statement I is incorrect but Statement II is correct.
- Both Statement I and Statement II are correct.
- Both Statement I and Statement II are incorrect.
- Statement I is correct but Statement II is incorrect.

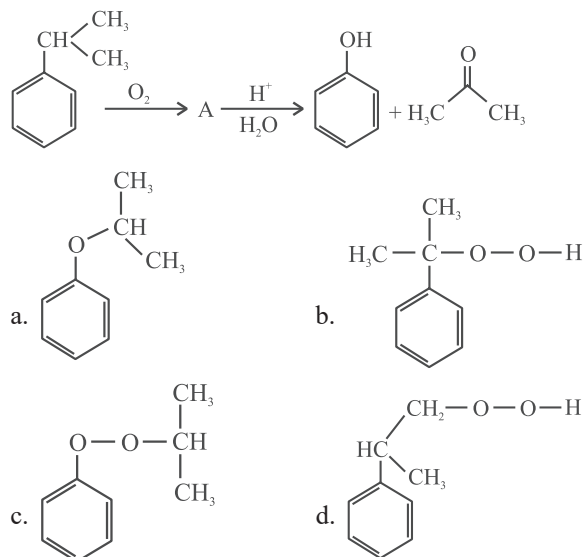
9. Which of the following substituted phenols is the strongest acid? (2020-Covid)



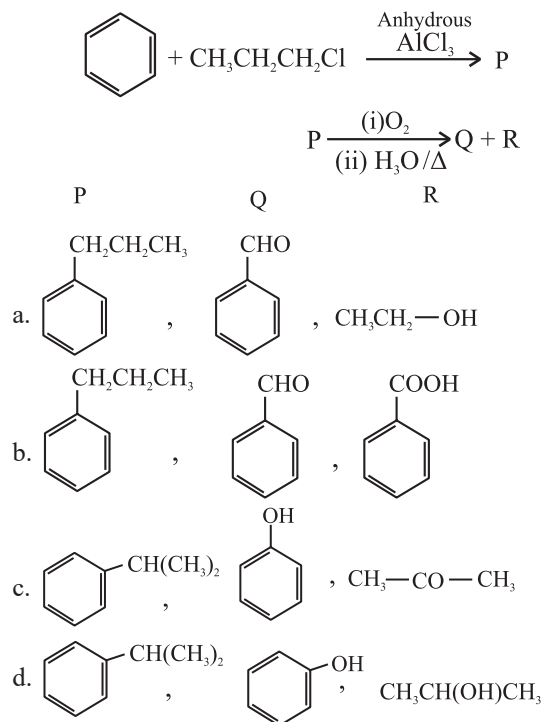
10. The compound that is most difficult to protonate is: (2019)



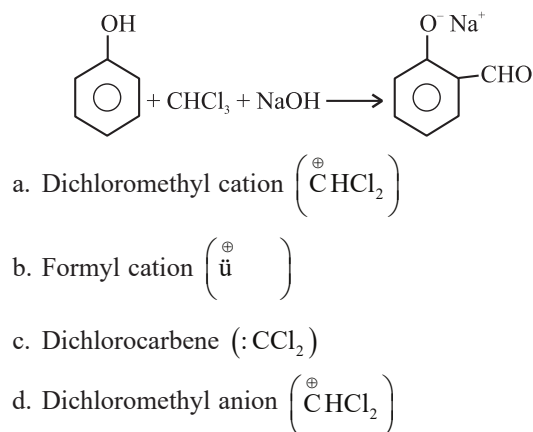
11. The structure of intermediate A in the following reaction, is (2019)



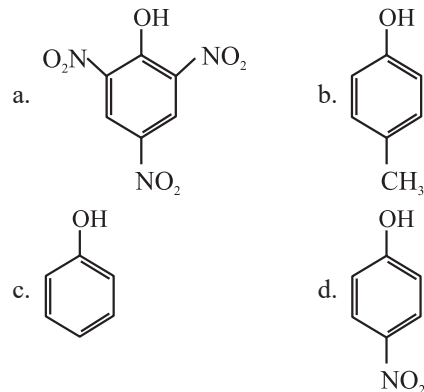
12. Identify the major products P, Q and R in the following sequence of reactions: (2018)



13. In the reaction, the electrophile involved is: (2018)



14. Which one is the most acidic compound? (2017-Delhi)



15. Reaction of phenol with chloroform in presence of dilute sodium hydroxide finally introduces which one of the following functional group? (2015 Re)

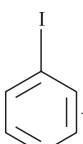
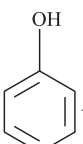
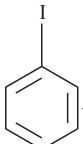
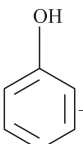
- CHO
- CH₂Cl
- COOH
- CHCl₂

16. Which of the following will not be soluble in sodium hydrogen carbonate? (2014)

- a. Benzoic acid b. o-Nitrophenol
c. Benzenesulphonic acid d. 2,4,6-trinitrophenol

Ethers

17. Anisole on cleavage with HI gives: (2020)

- a.  + CH₃OH b.  + C₂H₅I
c.  + C₂H₅OH d.  + CH₃I

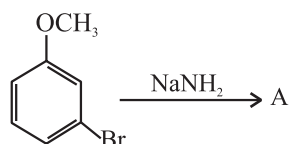
18. The compound A on treatment with Na gives B, and with PCl₅ gives C. B and C react together to give diethyl ether. A, B and C are in the order: (2018)

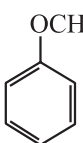
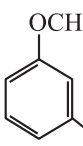
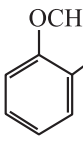
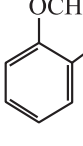
- a. C₂H₅OH, C₂H₆, C₂H₅Cl
b. C₂H₅OH, C₂H₅Cl, C₂H₅ONa
c. C₂H₅OH, C₂H₅ONa, C₂H₅Cl
d. C₂H₅Cl, C₂H₆, C₂H₅OH

19. The heating of phenyl-methyl ethers with HI produces. (2017-Delhi)

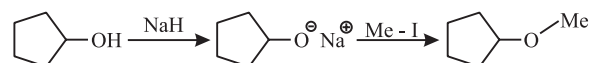
- a. Benzene b. Ethyl chlorides
c. Iodobenzene d. Phenol

20. Identify A and predict the type of reaction: (2017-Delhi)



- a.  and cine substitution reaction
b.  and substitution reaction
c.  and elimination addition reaction
d.  and cine substitution reaction

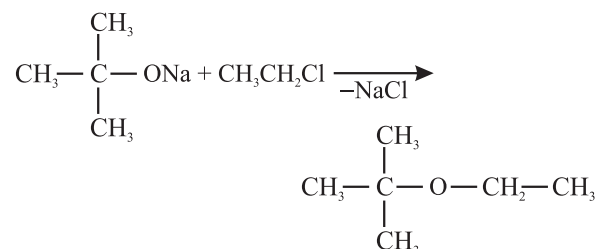
21. The reaction:



can be classified as? (2016 - I)

- a. Williamson alcohol synthesis reaction
b. Williamson ether synthesis reaction
c. Alcohol formation reaction
d. Dehydration reaction

22. The reaction



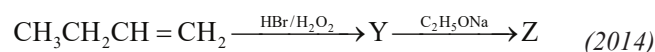
is called: (2015)

- a. Williamson continuous esterification process
b. Etard reaction
c. Gatterman - Koch reaction
d. Williamson synthesis

23. Among the following sets of reactants which one produces anisole? (2014)

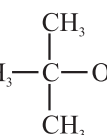
- a. C₆H₅OH; NaOH; CH₃I
b. C₆H₅OH; neutral FeCl₃
c. C₆H₅ - CH₃; CH₃COCl; AlCl₃
d. CH₃CHO; RMgX

24. Identify Z in the sequence of reactions:



- a. (CH₃)₂CH - O - CH₂CH₃
b. CH₃(CH₂)₄ - O - CH₃
c. CH₃CH₂ - CH(CH₃) - O - CH₂CH₃
d. CH₃ - (CH₂)₃ - O - CH₂CH₃

25. Among the following ethers, which one will produce methyl alcohol on treatment with hot concentrated HI? (2013)

- a. CH₃ - CH₂ - CH₂ - CH₂ - O - CH₃
b. 
c. 
d. 

Answer Key

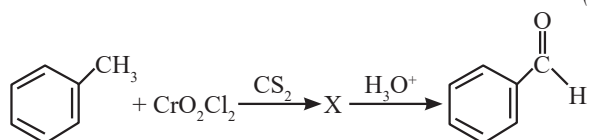
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
d	c	c	b	d	c	a	d	d	d	b	c	c	a	a	b	d
18	19	20	21	22	23	24	25									
c	d	b	b	d	a	d	c									

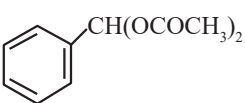
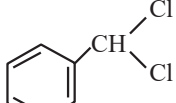
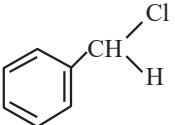
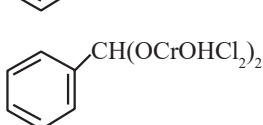
CHAPTER 18

Aldehydes, Ketones and Carboxylic Acids

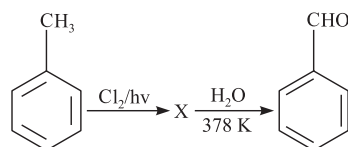
Preparation of Aldehydes and Ketones

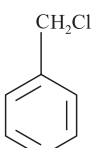
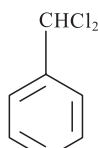
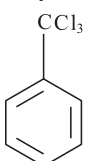
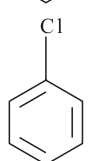
1. The intermediate compound 'X' in the following chemical reaction is: (2021)



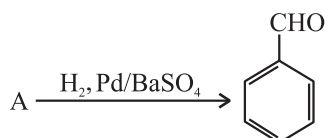
- a.  b. 
c.  d. 

2. Identify compound X in the following sequence of reactions: (2020)



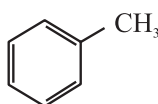
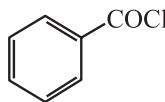
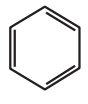
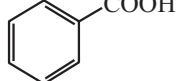
- a.  b. 
c.  d. 

3. Identify compound (A) in the following reaction: (2020-Covid)



- a. Toluene b. Acetophenone
c. Benzoic acid d. Benzoyl chloride

4. Reaction by which Benzaldehyde cannot be prepared: (2013)

- a.  + CrO_2Cl_2 in CS_2 followed by H_3O^+
b.  + H_2 in presence of Pd-BaSO_4
c.  + $\text{CO} + \text{HCl}$ in presence of anhydrous AlCl_3
d.  + Zn/Hg and concentration HCl

Physical Properties and Chemical Reactions of Aldehydes and Ketones

5. Given below are two statements: (2022)

Statements: I The boiling points of aldehydes and ketones are higher than hydrocarbons of comparable molecular mass because of weak molecular association in aldehydes and ketones due to dipole - dipole interactions.

Statements: II : The boiling points of aldehydes and ketones are lower than the alcohols of similar molecular mass due to the absence of H-bonding.

In the light of the above statements, choose the most appropriate answer from the options given below

- a. Statements I is incorrect but Statements II is correct.
b. Both Statements I and Statement II are correct.
c. Both Statements I and Statements II are incorrect.
d. Statements I is correct but Statements II is incorrect.

6. Match List-I with List-II. (2022)

List-I

(Products formed)

- (a) Cyanohydrin
(b) Acetal
(c) Schiff's base
(d) Oxime

List-II

(Reaction of carbonyl compound with)

- (i) NH_2OH
(ii) RNH_2
(iii) alcohol
(iv) HCN

a. (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
b. (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
c. (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
d. (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)

a. 

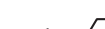
b. 

c. 

d. 

- Cannizzaro's reaction
- Cross Cannizzaro's reaction
- Cross Aldol condensation
- Aldol condensation

a. 

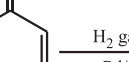
b. 

c. 

d. 

X $\xrightarrow{\text{Cu}/573 \text{ K}}$ A $\xrightarrow[\text{-OH, } \Delta]{[\text{Ag}(\text{NH}_3)_2]^+}$ S Silver mirror
 $(\text{C}_2\text{H}_6\text{O})$
 observed
 Y
 Z $\text{NH}_2-\text{NH}-\overset{\text{O}}{\parallel}\text{C}-\text{NH}_2$

- A-Ethanol, X-Acetaldehyde, Y-Butanone, Z-Hydrazine
- A-Methoxymethane, X-Ethanoic acid, Y-Acetate ion, Z-hydrazine
- A-Methoxymethane, X-Ethanol, Y-Ethanoic acid, Z-Semicarbazide
- A-Ethanal, X-Ethanol, Y-But-2-enal, Z-Semicarbazone



$$\text{Cyclohex-2-en-1-one} \xrightarrow[\text{Pd/carbon, ethanol}]{\text{H}_2 \text{ gas, 1 atmosphere}} \text{A}$$

a. 
 b. 

c. 
 d. 

a. Aromatic acid b. Schiff base
c. Ketone d. Carboxylic acid

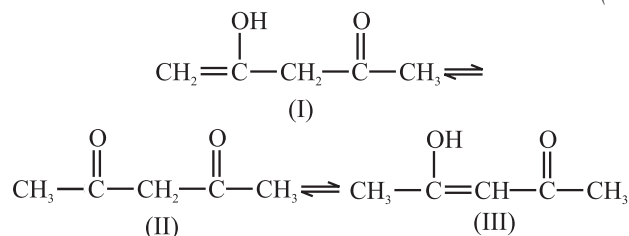
a. MnO_2 b. Aluminium isopropoxide
c. Acetone d. Ozone

- A carbonyl compound with a hydrogen atom on its α -carbon rapidly equilibrates with its corresponding enol and this process is known as carbonylation
- A carbonyl compound with a hydrogen atom on its α -carbon rapidly equilibrates with its corresponding enol and this process is known as keto-enol tautomerism
- A carbonyl compound with a hydrogen atom on its α -carbon never equilibrates with its corresponding enol
- A carbonyl compound with a hydrogen atom on its α -carbon rapidly equilibrates with its corresponding enol and this process is known as aldehyde-ketone equilibration

- Sodium hydrogen sulphite
- A Grignard reagent
- Hydrazine in presence of feebly acidic solution
- Hydrocyanic acid

a. 2-pentanone b. 3-pentanone
c. n-pentyl alcohol d. Pentanal

17. The order of stability of the following tautomeric compounds is: (2013)



- a. II > I > III b. II > III > I
c. I > II > III d. III > II > I

Methods of Preparation of Carboxylic Acids

18. $\text{RMgX} + \text{CO}_2 \xrightarrow[\text{ether}]{\text{dry}} \text{Y} \xrightarrow{\text{H}_3\text{O}^+} \text{RCOOH}$ (2022)

What is Y in the above reaction?

- a. $(\text{RCOO})_2\text{Mg}$ b. $\text{RCOO-Mg}^+\text{X}$
c. $\text{R}_3\text{CO-Mg}^+\text{X}$ d. RCOO-X^+

Physical Properties and Chemical Reactions of Carboxylic Acids

19. $\text{CH}_3\text{CH}_2\text{COO}^-\text{Na}^+ \xrightarrow[\text{Heat}]{\text{NaOH, ?}} \text{CH}_3\text{CH}_3 + \text{Na}_2\text{CO}_3$.

Consider the above reaction and identify the missing reagent/chemical. (2021)

- a. Red Phosphorus b. CaO
c. DIBAL-H d. B_2H_6

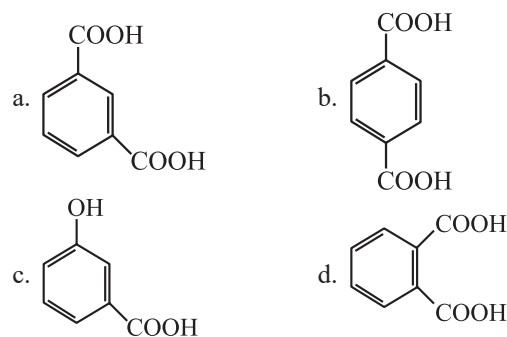
20. Match List-I with List-II. (2021)

List-I		List-II	
(A)	 $\xrightarrow[\text{AlCl}_3/\text{CuCl}]{\text{CO, HCl, Anhyd.}}$	(i)	Hell-Volhard-Zelinsky reaction
(B)	$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3 + \text{NaOX} \rightarrow$	(ii)	Gattermann-Koch reaction
(C)	$\text{R}-\text{CH}_2-\text{OH} + \text{R}'\text{COOH} \xrightarrow{\text{Conc. H}_2\text{SO}_4}$	(iii)	Haloform reaction
(D)	$\text{R}-\text{CH}_2\text{COOH} \xrightarrow[\text{(ii) H}_2\text{O}]{\text{(i) X}_2/\text{Red P}}$	(iv)	Esterification

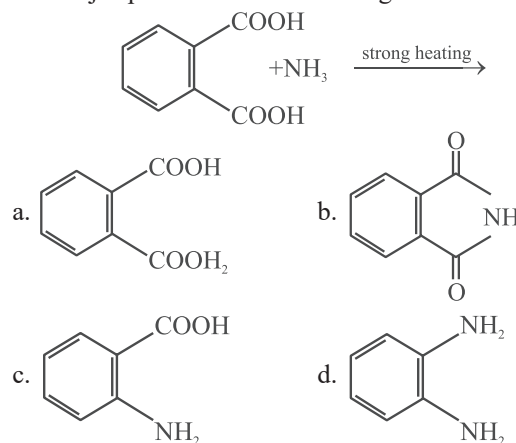
Choose the correct answer from the options given below.

- a. A-iii B-ii C-i D-iv
b. A-i B-iv C-iii D-ii
c. A-ii B-iii C-iv D-i
d. A-iv B-i C-ii D-iii

21. Which of the following acid will form an (i) Anhydride on heating and (ii) Acid imide on strong heating with ammonia? (2020-Covid)



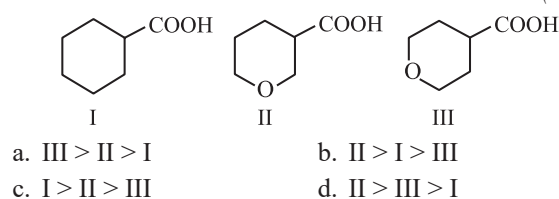
22. The major product of the following reaction is: (2019)



23. Carboxylic acids have higher boiling points than aldehydes, ketones and even alcohols of comparable molecular mass. It is due to their: (2018)

- a. Formation of intramolecular H-bonding
b. Formation of carboxylate ion
c. Formation of intermolecular H-bonding
d. More extensive association of carboxylic acid via van der Waals force of attraction

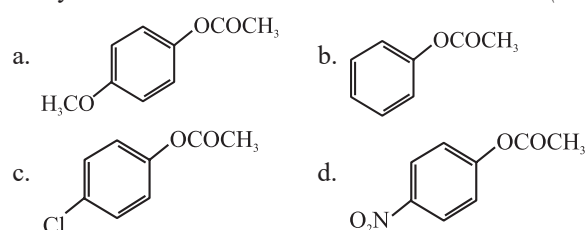
24. The correct order of strengths of the carboxylic Acids is: (2016 - II)



25. The oxidation of benzene by V_2O_5 in the presence of air produces: (2015 Re)

- a. Maleic anhydride b. Benzoic acid
c. Benzaldehyde d. Benzoic anhydride

26. Which one of the following esters gets hydrolysed most easily under alkaline conditions? (2015 Re)



Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
d	b	d	d	b	a	c	c	c	d	b	b	c	b	c	b	d
18	19	20	21	22	23	24	25	26								
b	b	c	d	b	c	d	a	d								

CHAPTER 19

Biomolecules

Carbohydrates

- Sucrose on hydrolysis gives: (2020)
 - α -D-Glucose + β -D-Glucose
 - α -D-Glucose + β -D-Fructose
 - α -D-Fructose + β -D-Fructose
 - β -D-Glucose + α -D-Fructose
- Which of the following statement is not true about glucose? (2020-Covid)
 - It contains five hydroxyl groups
 - It is a reducing sugar
 - It is an aldopentose
 - It is an aldohexose
- The difference between amylose and amylopectin is (2018)
 - Amylopectin have 1 $\rightarrow$ 4 α -linkage and 1 $\rightarrow$ 6 α -linkage
 - Amylose have 1 $\rightarrow$ 4 α -linkage and 1 $\rightarrow$ 6 β -linkage
 - Amylose is made up of glucose and galactose
 - Amylopectin have 1 $\rightarrow$ 4 α -linkage and 1 $\rightarrow$ 6 β -linkage
- The correct corresponding order of names of four aldoses with configuration given below: (2016 - II)

$$\begin{array}{c} \text{CHO} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{CH}_2\text{OH} \end{array}$$

$$\begin{array}{c} \text{CHO} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{CH}_2\text{OH} \end{array}$$

$$\begin{array}{c} \text{CHO} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{CH}_2\text{OH} \end{array}$$

$$\begin{array}{c} \text{CHO} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{CH}_2\text{OH} \end{array}$$

 - L-erythrose, L-threose, L-erythrose, D-threose
 - D-threose, D-erythrose, L-threose, L-erythrose
 - L-erythrose, L-threose, D-erythrose, D-threose
 - D-erythrose, D-threose, L-erythrose, L-threose
- Which one given below is a non-reducing sugar? (2016 - I)
 - Sucrose
 - Maltose
 - Lactose
 - Glucose
- D(+) glucose reacts with hydroxyl amine and yields an oxime. The structure of the oxime would be: (2014)

$$\begin{array}{c} \text{CH}=\text{NOH} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{CH}_2\text{OH} \end{array}$$

a.

$$\begin{array}{c} \text{CH}=\text{NOH} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{CH}_2\text{OH} \end{array}$$

b.

$$\begin{array}{c} \text{CH}=\text{NOH} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{CH}_2\text{OH} \end{array}$$

c.

$$\begin{array}{c} \text{CH}=\text{NOH} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{CH}_2\text{OH} \end{array}$$

d.

Proteins

- Which of the following is a basic amino acid? (2020)
 - Alanine
 - Tyrosine
 - Lysine
 - Serine
- The non-essential amino acid among the following is: (2019)
 - Valine
 - Leucine
 - Alanine
 - Lysine
- Which of the following compounds can form a zwitter ion? (2018)
 - Aniline
 - Acetanilide
 - Glycine
 - Benzoic acid
- In a protein molecule, various amino acids are linked together by: (2016 - I)
 - Dative bond
 - α -glycosidic bond
 - β -glycosidic bond
 - Peptide bond

Enzymes and Vitamins

11. The **incorrect** statement regarding enzymes is (2022)
- Enzymes are very specific for a particular reaction and substrate
 - Enzymes are biocatalysts
 - Like chemical catalysts enzymes reduce the activation energy of bio process.
 - Enzymes are polysaccharides.
12. The RBC deficiency is deficiency disease of: (2021)
- Vitamin B₆
 - Vitamin B₁
 - Vitamin B₂
 - Vitamin B₁₂
13. Deficiency of which vitamin causes osteomalacia? (2020-Covid)
- Vitamin D
 - Vitamin K
 - Vitamin E
 - Vitamin A

Nucleic Acids

14. The central dogma of molecular genetics states that the genetic information flows from: (2016 - II)

- Amino acids → Proteins → DNA
- DNA → Carbohydrates → Proteins
- DNA → RNA → Proteins
- DNA → RNA → Carbohydrates

15. The correct statement regarding RNA and DNA, respectively is: (2016 - I)
- The sugar component in RNA is 2'-deoxyribose and the sugar component in DNA is arabinose
 - The sugar component in RNA is arabinose and the sugar component in DNA is 2'-deoxyribose
 - The sugar component in RNA is ribose and the sugar component in DNA is 2'-deoxyribose
 - The sugar component in RNA is arabinose and the sugar component in DNA is ribose

Hormones

16. Which of the following hormones is produced under the conditions of stress which stimulate glycogenolysis in the liver of human beings? (2014)
- Thyroxin
 - Insulin
 - Adrenaline
 - Estradiol

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
b	c	a	d	a	c	c	c	c	d	d	d	a	c	c	c

CHAPTER 20

Haloalkanes and Haloarenes

Classification of Organohalogen Compounds, Nature of C-X Bond

1. The correct sequence of bond enthalpy of 'C-X' bond is: (2021)

- $\text{CH}_3 - \text{F} > \text{CH}_3 - \text{Cl} > \text{CH}_3 - \text{Br} > \text{CH}_3 - \text{I}$
- $\text{CH}_3 - \text{F} < \text{CH}_3 - \text{Cl} > \text{CH}_3 - \text{Br} > \text{CH}_3 - \text{I}$
- $\text{CH}_3 - \text{Cl} > \text{CH}_3 - \text{F} > \text{CH}_3 - \text{Br} > \text{CH}_3 - \text{I}$
- $\text{CH}_3 - \text{F} < \text{CH}_3 - \text{Cl} < \text{CH}_3 - \text{Br} < \text{CH}_3 - \text{I}$

Haloalkanes and Chemical Reactions of Haloalkanes

2. The major product formed in dehydrohalogenation reaction of 2-Bromo pentane is Pent-2-ene. This product formation is based on? (2021)

- Hund's rule
- Hofmann rule
- Huckel's rule
- Saytzeff's rule

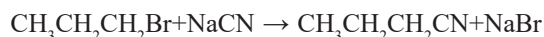
3. Elimination reaction of 2-Bromo-pentane to form pent-2-ene is (2020)

- β -Elimination reaction
 - Follows Zaitsev rule
 - Dehydrohalogenation reaction
 - Dehydration reaction
- (1), (3), (4)
 - (2), (3), (4)
 - (1), (2), (4)
 - (1), (2), (3)

4. An example of a sigma bonded organometallic compound is: (2017-Delhi)

- Cobaltocene
- Ruthenocene
- Grignard's
- Ferrocene

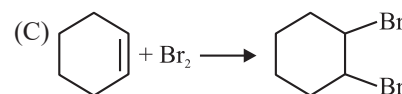
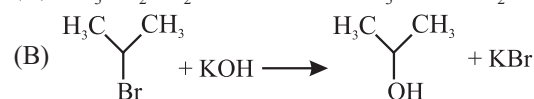
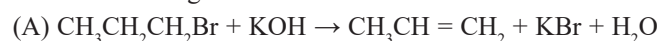
5. Consider the reaction



This reaction will be the fastest in: (2016 - II)

- N,N'-dimethylformamide (DMF)
- Water
- Ethanol
- Methanol

6. For the following reaction:



Which of the following statements is correct? (2016 - I)

- (A) is substitution, (B) and (C) are addition reactions.
 - (A) and (B) are elimination reactions and (C) is addition reaction.
 - (A) is elimination, (B) is substitution and (C) is addition reaction.
 - (A) is elimination, (B) and (C) are substitution reactions.
7. Which of the following reaction(s) can be used for the preparation of alkyl halides? (2015 Re)
- $\text{CH}_3\text{CH}_2\text{OH} + \text{HCl} \xrightarrow{\text{anh. ZnCl}_2}$
 - $\text{CH}_3\text{CH}_2\text{OH} + \text{HCl} \longrightarrow$
 - $(\text{CH}_3)_3\text{COH} + \text{HCl} \longrightarrow$
 - $(\text{CH}_3)_2\text{CHOH} + \text{HCl} \xrightarrow{\text{anh. ZnCl}_2}$
- (III) and (IV) only
 - (I), (III) and (IV) only
 - (I) and (II) only
 - (IV) only

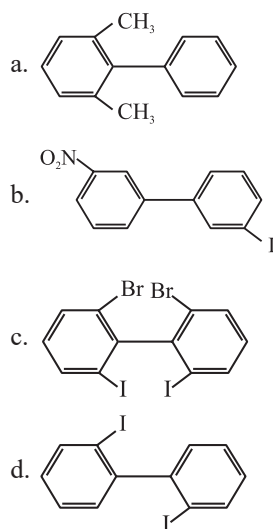
Stereochemical Aspects of Nucleophilic Substitution Reactions

8. The incorrect statement regarding chirality is: (2022)

- A racemic mixture shows zero optical rotation.
- $\text{S}_{\text{N}}1$ reaction yields 1 : 1 mixture of both enantiomers.
- The product obtained by $\text{S}_{\text{N}}2$ reaction of haloalkane having chirality at the reactive site shows inversion of configuration.
- Enantiomers are superimposable mirror images on each other.

9. Which of the following biphenyls is optically active?

(2016 - I)



10. Two possible stereo-structures of $\text{CH}_3\text{CHOHCOOH}$, which are optically active, are called:

(2015 Re)

- Mesomers
- Diastereomers
- Atropisomers
- Enantiomers

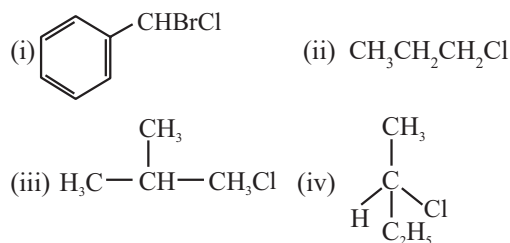
11. In a $\text{S}_{\text{N}}1$ reaction on chiral centres, there is:

(2015 Re)

- 100% inversion
- 100% racemisation
- Inversion more than retention leading to partial racemisation
- 100% retention

12. Which of the following compounds will undergo racemisation when solution of KOH hydrolyses?

(2014)

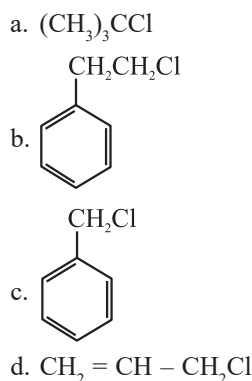


- (ii) and (iv)
- (iii) and (iv)
- (i) and (iv)
- (i) and (ii)

Haloarenes and Chemical Reactions of Haloarenes

13. Which of the following will NOT undergo $\text{S}_{\text{N}}1$ reaction with OH^- ?

(2020-Covid)

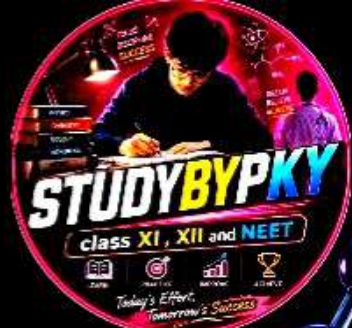


Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13
a	d	d	c	a	c	b	d	c	d	c	c	b

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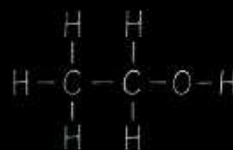
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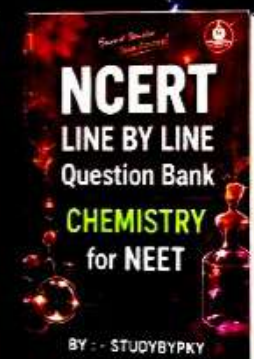
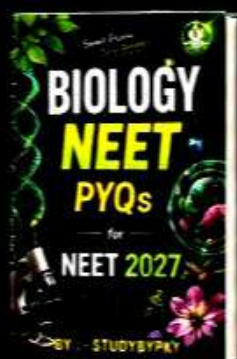
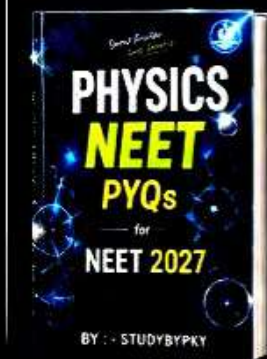
Chemistry
is not just about
reactions,
it's about
understanding
transformations
that change
the world.

$$E = h\nu$$

$$\Delta G = \Delta H - T\Delta S$$



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